



Libraries

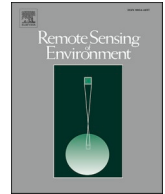
You have received this scanned document through Gelman Library's Interlibrary Loan Office.

If you notice any errors in its quality or otherwise, please contact us at ill@gwu.edu or 202-994-7128. This document will be removed in 30 days.

Thank you for using our services!

Notice: Warning Concerning Copyright Restrictions

The copyright law of the United States (Title 17, United States Code) governs the making of photocopies or other reproductions of copyrighted material. Under certain conditions specified in the law, libraries and archives are authorized to furnish a photocopy or other reproduction. One of these specific "fair use" conditions is that the photocopy or reproduction is not to be "used for any purpose other than private study, scholarship, or research." If a user makes a request for, or later uses, a photocopy or reproduction for purposes in excess of "fair use," that user may be liable for copyright infringement. No further reproduction and distribution of this copy is permitted by transmission or any other means.



Review

Transfer learning in environmental remote sensing

Yuchi Ma^{a,*}, Shuo Chen^b, Stefano Ermon^c, David B. Lobell^{a,*}^a Department of Earth System Science and Center on Food Security and the Environment, Stanford University, Stanford, CA 94305, USA^b Department of Earth, Atmospheric, and Planetary Sciences, Purdue University, West Lafayette, IN 47907, USA^c Department of Computer Science, Stanford University, Stanford, CA 94305, USA

ARTICLE INFO

Keywords:

Remote sensing
Domain shift
Transfer learning
Unsupervised domain adaptation
Multi-task learning
Few-shot learning
Self-supervised learning

ABSTRACT

Machine learning (ML) has proven to be a powerful tool for utilizing the rapidly increasing amounts of remote sensing data for environmental monitoring. Yet ML models often require a substantial amount of ground truth labels for training, and models trained using labeled data from one domain often demonstrate poor performance when directly applied to other domains. Transfer learning (TL) has emerged as a promising strategy to address domain shift and alleviate the need for labeled data. Here we provide the first systematic review of TL studies in environmental remote sensing. We start by defining the different forms of domain shift and then describe five commonly used TL techniques. We then present the results of a systematic search for peer-reviewed articles published between 2017 and 2022, which identified 1676 papers. Applications of TL in remote sensing have increased rapidly, with nearly 10 times more publications in 2022 than in 2017. Across seven categories of applications (land cover mapping, vegetation monitoring, soil property estimation, crop yield prediction, biodiversity monitoring, water resources management, and natural disaster management) we identify several recent successes of TL as well as some remaining research gaps. Finally, we highlight the need to organize benchmark datasets explicitly for TL in remote sensing for model evaluation. We also discuss potential research directions for TL studies in environmental remote sensing, such as realizing scale transfer, improving model interpretability, and leveraging foundation models for remote sensing tasks.

1. Introduction

In recent decades, significant advances in spaceborne platforms, airborne platforms, and sensors have dramatically increased the quantity and quality of environmental observations from remote sensing. Landsat, a series of Earth-observing satellites jointly operated by NASA and the U.S. Geological Survey, has provided multispectral and thermal imagery with global coverage of the Earth's surface for more than 50 years. Its latest generation, Landsat 9, was launched successfully in Sep 2021 and provides 30 m imagery at 10 spectral and thermal bands with a 16-day revisit time. There are also spaceborne platforms observing the Earth's surface with higher temporal or spatial resolutions, such as the Terra and Aqua satellites by NASA, Sentinel-2 by the European Space Agency (ESA), and Planet's PlanetScope and Skysat sensors. Imagery from these and other satellites has been widely used for environmental studies (Marshall et al., 2022; Peng et al., 2019; Wang et al., 2019). In addition to optical imagery, there emerge other sensing techniques like Synthetic-aperture radar (SAR) imaging (ESA's Sentinel-1) and Light Detection and Ranging (LiDAR) Scanning (NASA's Global Ecosystem

Dynamics Investigation (GEDI) mission), which complements information contained in optical imagery. SAR can penetrate through clouds and operate effectively in all weather conditions, which makes it valuable for forest monitoring and disaster management (Liu et al., 2021; Zhao et al., 2020). LiDAR can capture three-dimensional data and has been used for crop mapping and topographic mapping (Di Tommaso et al., 2023; Shan and Toth, 2018). More recently, there has been increasing usage of airborne platforms (aircraft and unmanned aerial vehicles (UAVs)) equipped with remote sensing instruments to capture high-resolution and specialized data in specific regions. Such platforms provide high flexibility and enable real-time data acquisition. For example, UAVs mounted with high-resolution sensors have found extensive applications in precision agriculture, where they play a crucial role in overseeing crop development and assisting in decision-making for farm management (Maimaitijiang et al., 2020; Sun et al., 2022). In summary, these platforms and sensing technology have revolutionized the way people monitor the Earth's surface and offered approaches to understanding environmental changes that may not be apparent from the ground.

With this increasing availability of remote sensing data, machine

* Corresponding authors at: Department of Earth Systems Science, Stanford University, 473 Via Ortega Dr., Stanford, CA 94305, USA.

E-mail addresses: yuchima8@gmail.com (Y. Ma), dlobell@stanford.edu (D.B. Lobell).

<https://doi.org/10.1016/j.rse.2023.113924>

Received 18 August 2023; Received in revised form 25 October 2023; Accepted 19 November 2023

Available online 28 November 2023

0034-4257/© 2023 Elsevier Inc. All rights reserved.

learning (ML) and especially deep learning (DL) have emerged to create new opportunities to quantify and understand changes in the environment that arise from human activity and natural disasters (Chen et al., 2022a; Liakos et al., 2018). Compared to visual inspection and process-based models, ML methods can be more scalable because they do not require expert domain knowledge or prior information on environmental parameters. Recently, DL models such as recurrent neural networks (RNN), convolutional neural networks (CNN), and transformers have been proposed to handle various data structures and dependencies. Specifically, RNNs have a looping structure that allows them to capture the dependencies in sequential data. CNNs employ convolutional layers that use a set of kernels to learn features like edges, textures, and patterns within the input images. Transformers apply the self-attention mechanism to weigh the contribution of different parts of the input data to the final predictions, enabling it to capture complex relationships and global dependencies in data (Vaswani et al., 2017). These models have been widely used in studies of environmental remote sensing, including land cover mapping (Rußwurm and Körner, 2020; Wang et al., 2021c), crop yield prediction (Ma et al., 2021a), and soil property estimation (Hengl et al., 2017). Moreover, there is an increasing number of products generated based on ML and remote sensing data for research and public use, such as the cropland layer (CDL) by the United States Department of Agriculture (Boryan et al., 2011).

Despite ongoing successes, ML models, especially DL models, commonly involve millions of parameters and need large amounts of labeled data for model training (Gadiraju and Vatsavai, 2020; Lu et al., 2023). For example, TempCNNs for the classification of satellite image time series has 2.5 M parameters, which require 1 M labeled time series for training (Pelletier et al., 2019). Similarly, the latest Vision Transform for satellite image time series contains 1.7 M parameters and needs more than 2 M labeled images for training (Tarasiou et al., 2023). Such labeled data are costly to collect, especially for studies in agriculture and forestry that involve labor-intensive field campaigns and destructive samplings. When insufficient labeled data is available, training reliable DL models is challenging and issues such as overfitting and underfitting are common. Moreover, the difference in data distributions arising from variations in data collection dates, spectral ranges, and geospatial environments (i.e., domain shift) (Michau and Fink, 2021) mean that DL models trained using labeled data from a domain with a surplus of labeled data (known as the “source domain”) often demonstrate poor performance when directly applied to a task in the domain of interest (known as the “target domain”). Domain shift therefore commonly limits the generalizability and transferability of DL models. For example, Gadiraju and Vatsavai (2020) directly applied a ResNet trained with natural images to classify crops in aerial images, which achieved an accuracy of 55.14%, much worse than 82.85% accuracy by the model trained with the aerial images. Similarly, Ma et al. (2021b) trained random forests and deep neural networks in one eco-region and used them to predict corn yield in the other region, observing a large decrease in the model performance when compared to the locally trained models.

Transfer learning (TL) is a ML technique that involves transferring knowledge gained from the source domain to finish the related tasks in the target domain (Pan and Yang, 2010). By leveraging the knowledge learned from the source domain, TL can help overcome the challenges of limited labeled data in the target domain. Recently, significant advancements have been made in TL techniques for various remote sensing applications (Bouchard et al., 2022; Ma et al., 2021b; Martini et al., 2021). Despite the success of TL, existing reviews of ML in remote sensing mainly focus on supervised learning methods and their applications in specific research areas. For example, Zhu et al. (2017) gave a review of DL in remote sensing with a focus on CNN-based supervised learning models. Tuia et al. (2016) focused on studies of unsupervised domain adaptation (UDA), which is a subfield of TL, for the classification of remote sensing data. Yuan et al. (2020) reviewed applications of DL methods to advance the environmental remote sensing process. Odebiri et al. (2021) present a brief review of DL models in remote sensing of soil

organic carbon estimation, while Liu and Wang (2021) focused on plant diseases and pest detection. A few of these reviews have mentioned TL as a promising strategy for overcoming small training datasets, yet none have provided a comprehensive view of the status of TL approaches.

To fill this gap, this paper reviews the use of TL in environmental remote sensing from the perspectives of both methodology and applications. In Section 2, we start by providing the definition and categorization of domain shift, followed by the setting and framework of TL. In Section 3, we describe five widely used TL techniques, which differ according to the availability of labeled data and training strategies. In Section 4, we review representative TL studies in seven categories of applications within the field of environmental remote sensing: land cover mapping, vegetation monitoring, soil property estimation, crop yield prediction, biodiversity monitoring, water resources management, and natural disaster management. In Section 5, we discuss gaps in current research and corresponding future research directions. Finally, we summarize the conclusions in Section 6.

2. Domain shift and transfer learning

Given a learning task T with training samples $(\mathbf{x}, \mathbf{y})$ drawn from a joint data distribution $p(\mathbf{x}, \mathbf{y})$ and the corresponding loss function (i.e., cross entropy for classification), ML and DL models are trained to associate input feature variables $\mathbf{x} \in X$ with response variables $\mathbf{y} \in Y$ by minimizing the empirical risk calculated by averaging the loss function on the training set, in which X and Y denote the input feature space and the output label space, respectively. In the supervised learning setting, the model learns to map the input data $\mathbf{x}$ to the label data $\mathbf{y}$ through a process of adjusting its internal parameters until it achieves the best possible accuracy on the training data (i.e., minimizing a loss function and optional regularization terms). Optimally, the trained model can generalize well to new data, and fulfill the task on data that the model has not been trained on.

Suppose there is a target domain with a learning task T_t . Let $X_t = \{\mathbf{x}_i\}_{i=1}^{N_t}$ and $Y_t = \{\mathbf{y}_i\}_{i=1}^{N_t}$ denote the feature variables and the corresponding data labels drawn from $p_t(\mathbf{x}, \mathbf{y})$. N_t denotes the number of data samples. If enough labeled samples are available (i.e., N_t is quite large), a ML model can be trained from scratch using supervised learning. However, labeled data may be scarce or expensive to acquire in the domain of interest. When the amount of labeled training data is limited, the model might start memorizing the data instead of generalizing from it, causing the issue of overfitting (Oikonomidis et al., 2022). Furthermore, when no labeled target sample is available, ML models cannot be trained from scratch in the target domain using supervised learning. Consequently, it is challenging to build effective and reliable ML models in both scenarios.

Fortunately, there may exist substantial amounts of labeled data as $X_s = \{\mathbf{x}_i\}_{i=1}^{N_s}$ and $Y_s = \{\mathbf{y}_i\}_{i=1}^{N_s}$ drawn from $p_s(\mathbf{x}, \mathbf{y})$ in another domain (known as the “source domain”). These labeled data are used for the same or relevant learning task T_s as the learning task T_t in the target domain. Nonetheless, due to differences between the data distributions in the two domains (i.e., $p_s(\mathbf{x}, \mathbf{y}) \neq p_t(\mathbf{x}, \mathbf{y})$), directly applying the ML model trained in the source domain to the target domain may lead to a substantial decrease in its performance. The differences between the data distributions in the source and target domains, known as domain shift, can render the transferred knowledge unhelpful or detrimental to the learning task in the target domain. According to Bayes’ theorem, the joint distribution can be decomposed into several components: $p(\mathbf{x}, \mathbf{y}) = p(\mathbf{x}|\mathbf{y})p(\mathbf{y}) = p(\mathbf{y}|\mathbf{x})p(\mathbf{x})$. Depending on the specific part where the distribution shift occurs, domain shift can be categorized into three types, including covariate shift, prior shift, and concept shift (Table 1) (Kouw and Loog, 2018; Nikzad-Langerodi and Andries, 2021; Quinero-Candela et al., 2008; Tuia et al., 2016).

Covariate shift: in the case of covariate shift, the marginal distributions in the source domain and target domain are different, i.e.,

Table 1
Description of different types of domain shift.

Type	Formal definition	Description	Example
Covariate shift	$p_s(\mathbf{x}) \neq p_t(\mathbf{x})$ $p_s(\mathbf{y} \mathbf{x}) \approx p_t(\mathbf{y} \mathbf{x})$	The marginal distributions in the feature spaces of the source and target domains are different.	Different sensors, such as Landsat-8 and Sentinel-2, have differences in their spatial resolutions and spectral ranges, causing shifts in reflectance values. The ImageNet dataset has different label spaces as downstream tasks, such as crop classification, in the target domain.
Prior shift	$p_s(\mathbf{y}) \neq p_t(\mathbf{y})$ $p_s(\mathbf{x} \mathbf{y}) \approx p_t(\mathbf{x} \mathbf{y})$	The prior distributions in the label space of the source and target domains are different	The same spectral reflectance value may correspond to distinct soil moisture values in different regions due to changes in soil textures and climates.
Concept shift	$p_s(\mathbf{x}) \approx p_t(\mathbf{x})$ $p_s(\mathbf{y} \mathbf{x}) \neq p_t(\mathbf{y} \mathbf{x})$	The marginal distributions remain similar, but the posterior distributions of the response given the inputs are different in the source and target domains	

$p_s(\mathbf{x}) \neq p_t(\mathbf{x})$, while the posterior distributions of $\mathbf{y}$ given $\mathbf{x}$ remain similar between the domains, i.e., $p_s(\mathbf{y}|\mathbf{x}) \approx p_t(\mathbf{y}|\mathbf{x})$ (Fig. 1). Covariate shift occurs when data in the same spectral bands are collected by different sensors, in distant geographic regions, or at different time windows. For instance, Landsat-8 and Sentinel-2 both provide observations in visible and near-infrared bands. However, variations in central wavelengths, bandwidths, and spatial resolutions between their sensors can lead to covariate shift between the Landsat-8 data and the Sentinel-2 data even when observing the same landcover (Wu et al., 2022). Similarly, when using UAV imagery for plant disease identification, covariate shift is expected to occur when data are collected during different time windows of the day due to changes in illumination conditions (Liu and Wang, 2021).

Prior shift: in the case of prior shift, the conditional distributions have high similarity but the prior distributions of the label space in the source domain and target domain are different, i.e., $p_s(\mathbf{y}|\mathbf{x}) \approx p_t(\mathbf{y}|\mathbf{x})$ and $p_s(\mathbf{y}) \neq p_t(\mathbf{y})$ (Fig. 2). Prior shift happens when the label spaces are different in the source and target domains. For instance, in landcover classification, the source domain dataset normally encompasses a vast area containing numerous landcover classes, each occurring equally often. In contrast, the target domain might be limited to a smaller geographical region characterized by a very different distribution of labels. As a result, we anticipate a significant prior shift between the two domains.

Concept shift: in concept shift the relationship between the feature variables and response variables is different in the source and target domains. Specifically, the marginal distributions of the covariates are

the same, but the conditional distributions are different, i.e., $p_s(\mathbf{x}) \approx p_t(\mathbf{x})$ and $p_s(\mathbf{y}|\mathbf{x}) \neq p_t(\mathbf{y}|\mathbf{x})$ (Fig. 3). Conditional shift can happen when remote sensing observations fail to capture influencing factors. For example, corn fields in two countries can have distinct yields even if they have similar reflectance, because factors such as irrigation or pest control may influence crop yield but are not fully captured by remote sensing observations, leading to varying relationships between the feature variables and response variables. Another example is soil moisture estimation, in which the same spectral reflectance value may correspond to distinct soil moisture values in different regions due to the changes in soil textures and climates.

To address domain shift and improve the transferability of ML models, TL has been proposed to transfer knowledge from a data-abundant source domain to the data-scarce target domain. The idea is that the knowledge and feature representations learned by the model in the source domain can be transferred to finish the task in the target domain, reducing the amount of data required and improving the accuracy and efficiency of the model in the target domain. Diverse TL techniques have been developed based on the characteristics of different types of domain shift and the accessibility of labeled data. The following section elaborates on these techniques.

3. A categorization of transfer learning techniques

To date, several TL techniques have been proposed to address various types of research questions in different scenarios. These TL techniques are summarized in Table 2 and described below.

3.1. Fine-tuning-based transfer learning

Fine-tuning-based TL (FTL) is a popular TL technique for deep neural networks (DNN), which have been widely used in a variety of remote sensing applications (Gadiraju and Vatsavai, 2020; Wang et al., 2018b). FTL involves first pre-training a model in D_s and fine-tuning its parameters for the task in D_t (Fig. 4). To learn robust and generalizable feature representations, it is critical to pre-train the model with a large, diverse dataset. Therefore, the number of source samples is normally larger than the number of target samples ($N_s > N_t$) and the learning tasks in the source and target domains are related.

However, when labeled samples in the target domain are insufficient, fine-tuning the entire neural network can lead to overfitting (Mehdipour Ghazi et al., 2017). Fortunately, DL models have layered architectures that learn distinct features at different layers. It has been found that the initial layers tend to capture generic features, while the later ones focus more on specific tasks (Yosinski et al., 2014). As such, some research froze the weights of earlier layers of a pre-trained DL architecture and customized the model to a given task by fine-tuning or retraining the last few layers (Abdalla et al., 2019).

One advantage of FTL is that it allows the adaptation of advanced pre-trained models to a variety of domains. Popular deep architectures, such as GoogLeNet (Szegedy et al., 2015), VGG (Simonyan and

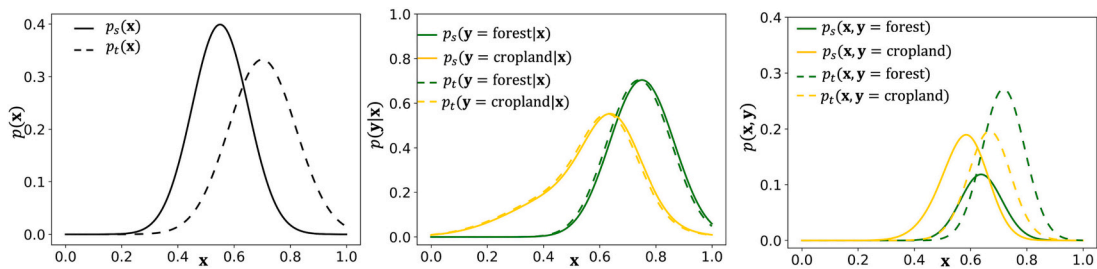


Fig. 1. An example of covariate shift when collecting satellite-derived NDVI ($\mathbf{x}$) $\in [0, 1]$ over the same region at two different times of the day for landcover classification $\mathbf{Y} = \{\text{forest, cropland}\}$: (Left) The marginal distributions are different due to changes in illumination conditions. (Middle) The posterior distributions are similar since the observation region is the same. (Right) The resulting joint distributions are shifted (Example inspired by (Kouw and Loog, 2018)).

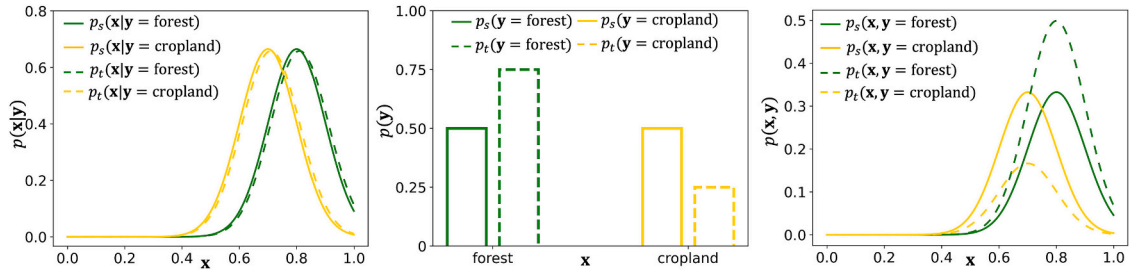


Fig. 2. An example of prior shift when collecting satellite-derived NDVI ($\mathbf{x}$) $\in [0, 1]$ in two nearby regions with different landcover distributions for landcover classification $Y = \{\text{forest, cropland}\}$: **(Left)** The conditional distributions have high similarity because two regions are nearby and share a similar environment. **(Middle)** The prior distributions are different as the source region has equal portions of forests and croplands but the target region has more forests. **(Right)** The resulting joint distributions are shifted (Example inspired by (Kouw and Loog, 2018)).

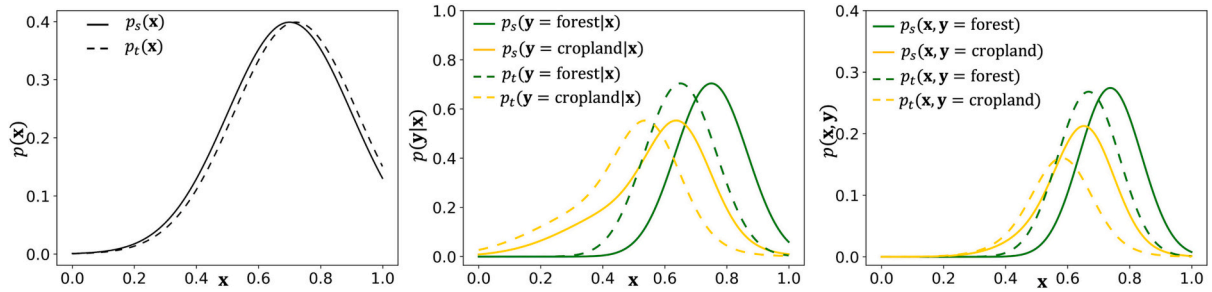


Fig. 3. An example of concept shift when collecting satellite-derived NDVI ($\mathbf{x}$) $\in [0, 1]$ in two distant regions with different water availability for landcover classification $Y = \{\text{forest, cropland}\}$: **(Left)** The marginal distributions have high similarity. **(Middle)** The posterior distribution in the target region shifts to the left since forests and crops tend to have smaller NDVI due to water shortage. **(Right)** The resulting joint distributions are shifted (Example inspired by (Kouw and Loog, 2018)).

Table 2

Settings of transfer learning techniques.

Transfer learning technique	Source label	Target label	Advantage
Fine-tuning-based transfer learning	Available	Available	- Faster convergence;
Multi-task learning	Available	Available	- High flexibility;
Few-shot learning	Available	Available	- Robust to overfitting;
Unsupervised domain adaptation	Available	Unavailable	- High data utilization;
Self-supervised learning	Unavailable	Available	- High generalizability;
			- High data efficiency;
			- No requirements for labeled target samples;
			- Utilization of both labeled and unlabeled data;
			- No requirements for labeled source samples;
			- Learn informative representations that can be used for various downstream tasks;

Zisserman, 2015), ResNet-50 (He et al., 2016), are trained on large datasets with millions of images, such as ImageNet (Deng et al., 2009). Through FTL, these models can be adapted for new tasks with a few data samples and still often achieve state-of-the-art results. In addition, since FTL uses labels in the target domain, it can address all types of domain shift. However, it is important to note that the effectiveness of FTL depends on the relatedness of the learning tasks in D_s and D_t . It is more likely to work when the pre-training task and the new task are in the same scope (e.g., image classification).

3.2. Multi-task learning

Unlike single-task learning which builds a model for one task, multi-task learning (MTL) is a TL technique where a single model is trained to perform multiple tasks simultaneously (Zhang and Yang, 2022). In MTL,

each learning task can be regarded as a domain. The goal of MTL is to maximize the model's generalizability and improve the model's performance on one or multiple tasks. All domains have a shared feature set while their learning tasks are different but related. The MTL model is trained to extract cross-domain features that are informative for both tasks so that the generalizability of the model is improved. For example, neural networks are commonly used for MTL, which are designed to have a shared set of intermediate layers and task-specific output layers for each domain (Fig. 5) (Feng et al., 2021; Nguyen et al., 2019). The weight-sharing intermediate layers act as the feature extractor to extract generalizable and robust features while independent task-specific layers are used to make predictions for each task.

In MTL, knowledge gained from each task is shared, leading to improved performance compared to training separate models for each single task. Training a single model for multiple tasks also acts as a form of regularization, which prevents overfitting to any single task and improves the generalization performance of the model. Moreover, MTL can improve the utilization of available data. When it is difficult to obtain large amounts of labeled data for each task individually, MTL can be used to leverage the availability of labeled data for multiple tasks. The effectiveness of MTL depends on the relatedness of the tasks, with tasks with high relatedness allowing the model to learn knowledge that is useful and transferable for each domain (Zhang and Yang, 2022). Normally, each domain in MTL is supposed to share the same feature space. Some recent studies on relatedness and redundancies across different domains have relaxed this limitation and made MTL work on multiple types of inputs (Zamir et al., 2019).

3.3. Few-shot learning

Few-shot learning (FSL) is a ML technique where a model is trained to generalize to a new task using very few labeled samples (Wang et al., 2020). In comparison with FTL, FSL has a strict requirement regarding the amount of labeled data samples. In FSL, various techniques are

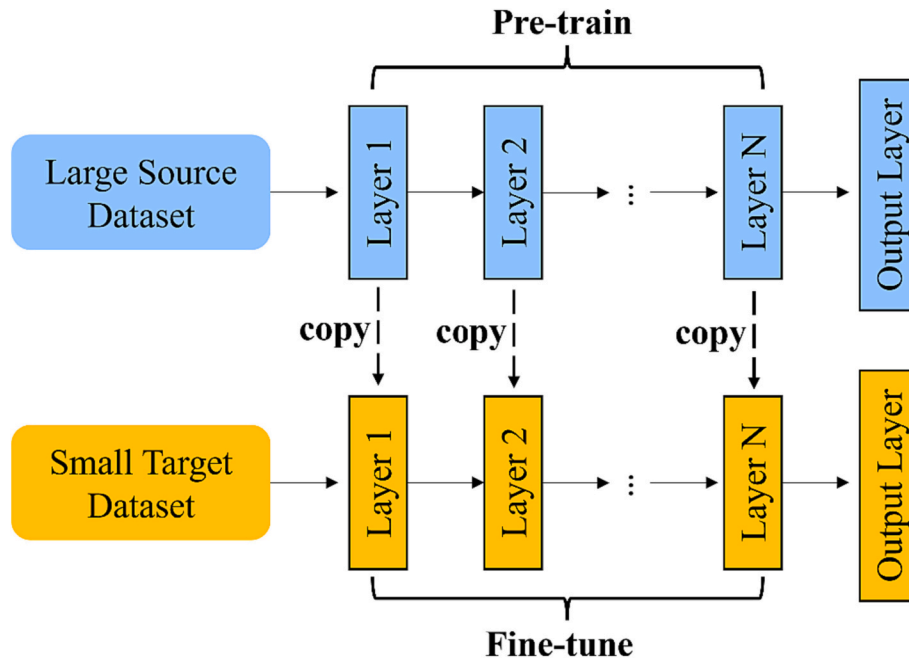


Fig. 4. The workflow of fine-tuning-based transfer learning (FTL). The model is first pre-trained with a large source dataset and then adapted to the target domain by fine-tuning it on a small target dataset.

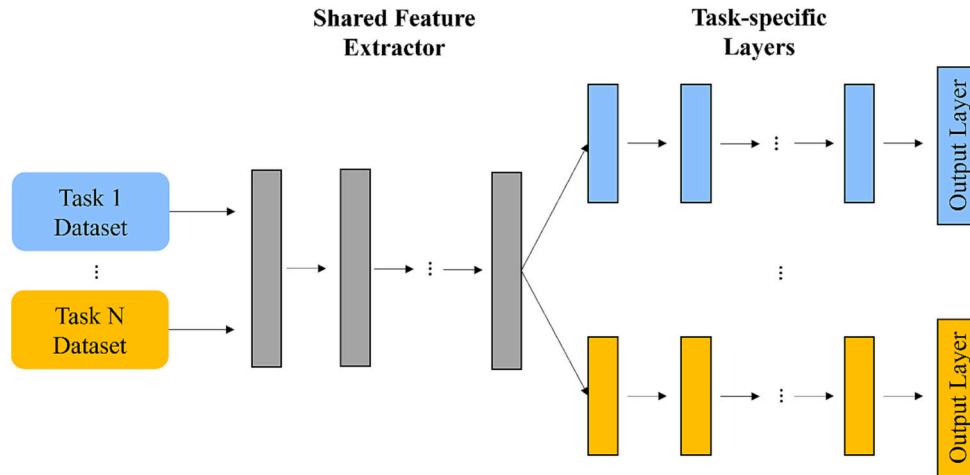


Fig. 5. The architecture of a multi-task learning (MTL) neural network. The model has a shared feature extractor to learn generalizable and robust features and task-specific layers to make predictions for each task.

employed, such as meta-learning and metric learning (Wang et al., 2020; Yang et al., 2022). Additionally, whereas FTL focuses on adapting a pre-trained model to a related task and may not generalize well to significantly different tasks or domains without additional fine-tuning or retraining, FSL is designed to handle diverse tasks and generalize more readily to new and unseen data.

The support set and the query set are two important components of FSL (Fig. 6). Specifically, the support set is a small set of labeled samples chosen to assist in learning new tasks. The query set is a set of unlabeled samples from the target domain for model evaluation. The query set typically contains examples from the same classes as the support set but is different from the ones in the support set. The query set is used to assess how well the model can generalize to new examples and classes, given the limited labeled examples in the support set. The support set and the query set in FSL can be seen as analogous to the source domain and the target domain, respectively.

FSL approaches can be broadly categorized into three types based on

the sample availability, including Few-shot (a few training samples per class), One-shot (a single training sample per class), and Zero-shot (zero training sample per class) (Lu et al., 2023). Depending on the learning strategies, FSL approaches can be further categorized into four types (Wang et al., 2020; Yang et al., 2022), including data-based FSL, metric-based FSL, optimization-based FSL, and model-based FSL. Data-based FSL approaches are proposed to solve the issue of missing labeled training data for the new task via data augmentation by generating synthetic data samples. This type of approach involves learning a generative model, such as a variational autoencoder or a generative adversarial network (GAN). Metric-based FSL is the most widely studied approach (Sung et al., 2018), in which the model aims at learning a metric or similarity function that can compare a query example to the support examples for a task, and make predictions based on the similarities (Fig. 6). Based on the similarity scores between the support and query sets, the model can draw inferences and make predictions for unseen query samples. Optimization-based FSL aims to learn an

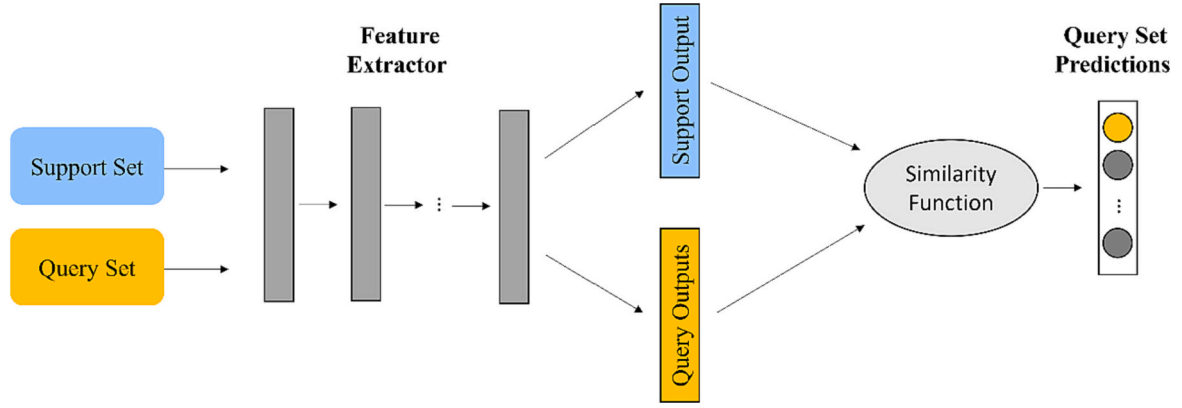


Fig. 6. The workflow of metrics-based few-shot learning (FSL). The model learns to make predictions for the query examples based on the similarities to the labeled examples in the support set.

appropriate initialization parameter and update the model parameters in a few steps without overfitting. Model-based FSL tries to realize rapid adaptation from the perspective of the model architecture. Examples include approaches based on external memory, in which additional memory modules are added to the model to store the characteristic information of the few unseen samples (Wang et al., 2020; Yang et al., 2022).

In summary, FSL enables models to quickly adapt and generalize to new domains with limited labeled data. FSL leverages prior knowledge from similar tasks, allowing them to learn new concepts efficiently. More importantly, FSL provides high flexibility and enables models to generalize well to unseen tasks, which makes FSL suitable for real-world applications where new data and tasks are frequently emerging. Correspondingly, FSL can be used to address prior shift and exhibits strong adaptability to new and unseen tasks.

3.4. Unsupervised domain adaptation

Unsupervised domain adaptation (UDA) addresses domain shift by aligning feature distributions in the source and target domains based on labeled source data and unlabeled target data (Fig. 7) (Tuia et al., 2016). In the setting of UDA, D_s and D_t have the same learning task and there are sufficient labeled samples in D_s while there are only unlabeled samples in D_t (Zhao et al., 2022b). UDA models commonly are designed to have a conjugated architecture with two objectives to respectively fulfill the main learning task and the task of reducing domain shift. The main objective is to learn a task-specific model based on the labeled samples from the source domain by minimizing the corresponding loss function, such as mean squared error (MSE) for regression (Sun et al.,

2022) and cross-entropy loss for classification (Kwak and Park, 2022). The other objective is to align the feature distributions between D_s and D_t by reducing domain shift.

Based on the feature alignment strategies, UDA approaches can be generally categorized into transformation-based methods, discrepancy-based methods, adversarial-based methods, and generative-based methods (Kouw and Loog, 2019; Wang and Deng, 2018). Transformation-based methods map instances from the source domain and the target domain into a subspace with transferable components. Representative examples include transfer component analysis (TCA) (Pan et al., 2011), which transforms source and target data into a common feature space by linearly combining their kernel features using a weight matrix. Discrepancy-based methods align source and target features by minimizing their distribution discrepancy such as Maximum Mean Discrepancy (Othman et al., 2017), Wasserstein Discrepancy (Shen et al., 2018), and Moment Discrepancy (Zellinger et al., 2017). However, discrepancy-based methods commonly involve calculations of discrepancy with high complexity (Chadha and Andreopoulos, 2019). Recently, adversarial-based models have become increasingly popular due to their simplicity in training and success in minimizing domain shift. Adversarial-based models are trained via adversarial learning until the source and target domains are aligned. Domain adversarial neural network (DANN) is among the most popular adversarial-based UDA models (Fig. 8), in which a domain classifier is employed to measure domain shift and a feature extractor is trained adversarially against the domain classifier to extract domain-invariant features (Ganin et al., 2017). Moreover, generative-based methods are based on GAN to realize style transfer between the source domain and the target domain (Bellocchio et al., 2020). These approaches typically employ a generator to

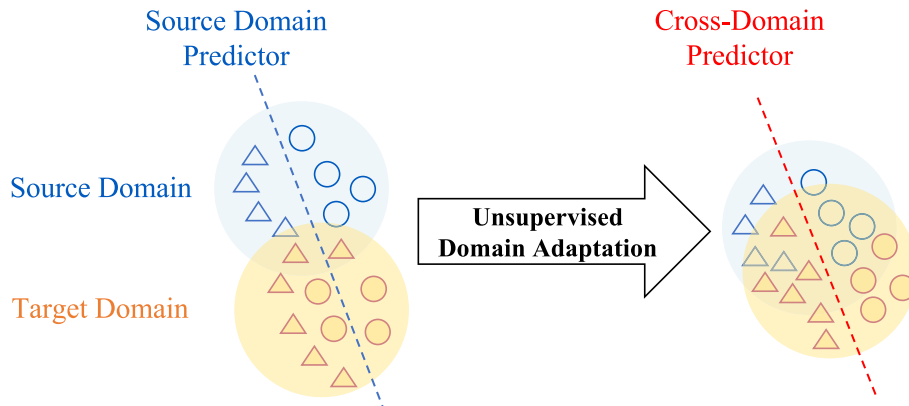


Fig. 7. The conceptual example of unsupervised domain adaptation (UDA). Before UDA, the predictor trained on the source domain makes erroneous predictions in the target domain. After UDA, the source and target domains are aligned, and the cross-domain predictor can make accurate predictions in both domains.

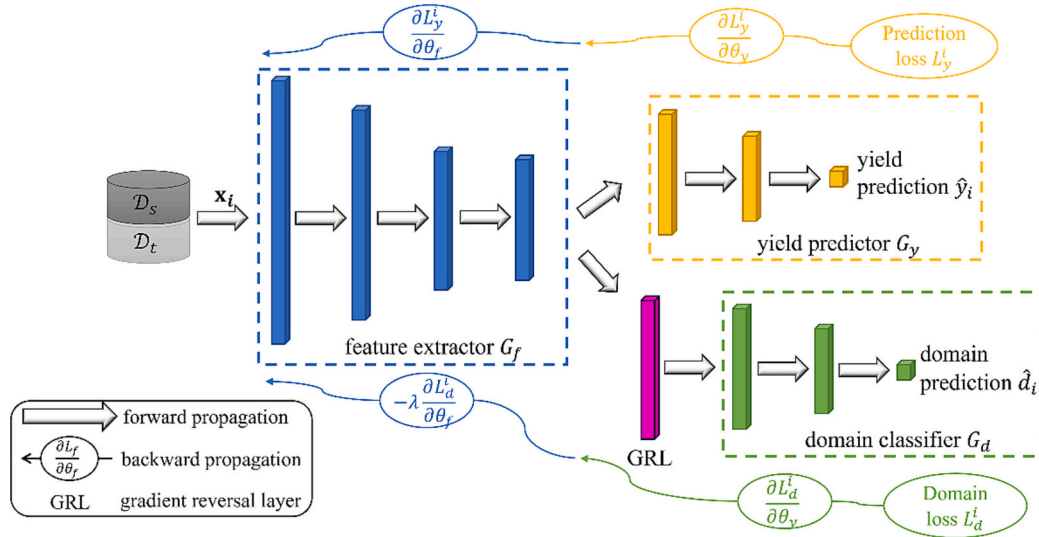


Fig. 8. The domain adversarial neural network (DANN) for crop yield prediction (Ma et al., 2021b). Feature vectors from the source domain $\mathcal{D}_s$ and the target domain $\mathcal{D}_t$ are projected into the cross-domain subspace with reduced domain shift.

reduce domain shift by transforming the source data to match the style of the target domain.

In essence, UDA reduces domain shift by aligning the feature distributions in the source and target domains. It enables ML and DL models to be more generalizable and transferable across different domains without the need for labeled data samples from the target domain. UDA methods utilize both labeled and unlabeled data, leading to improved

model performance on the target task. Most UDA approaches are designed to reduce covariate shift. Some variants of UDA, such as partial domain adaptation (PDA), can address prior shift by partially aligning the source and target domains in the shared label spaces (Gu et al., 2021; Zhang et al., 2018). There are also UDA methods proposed to reduce concept shift, such as maximum classification discrepancy (MCD) (Saito et al., 2018).

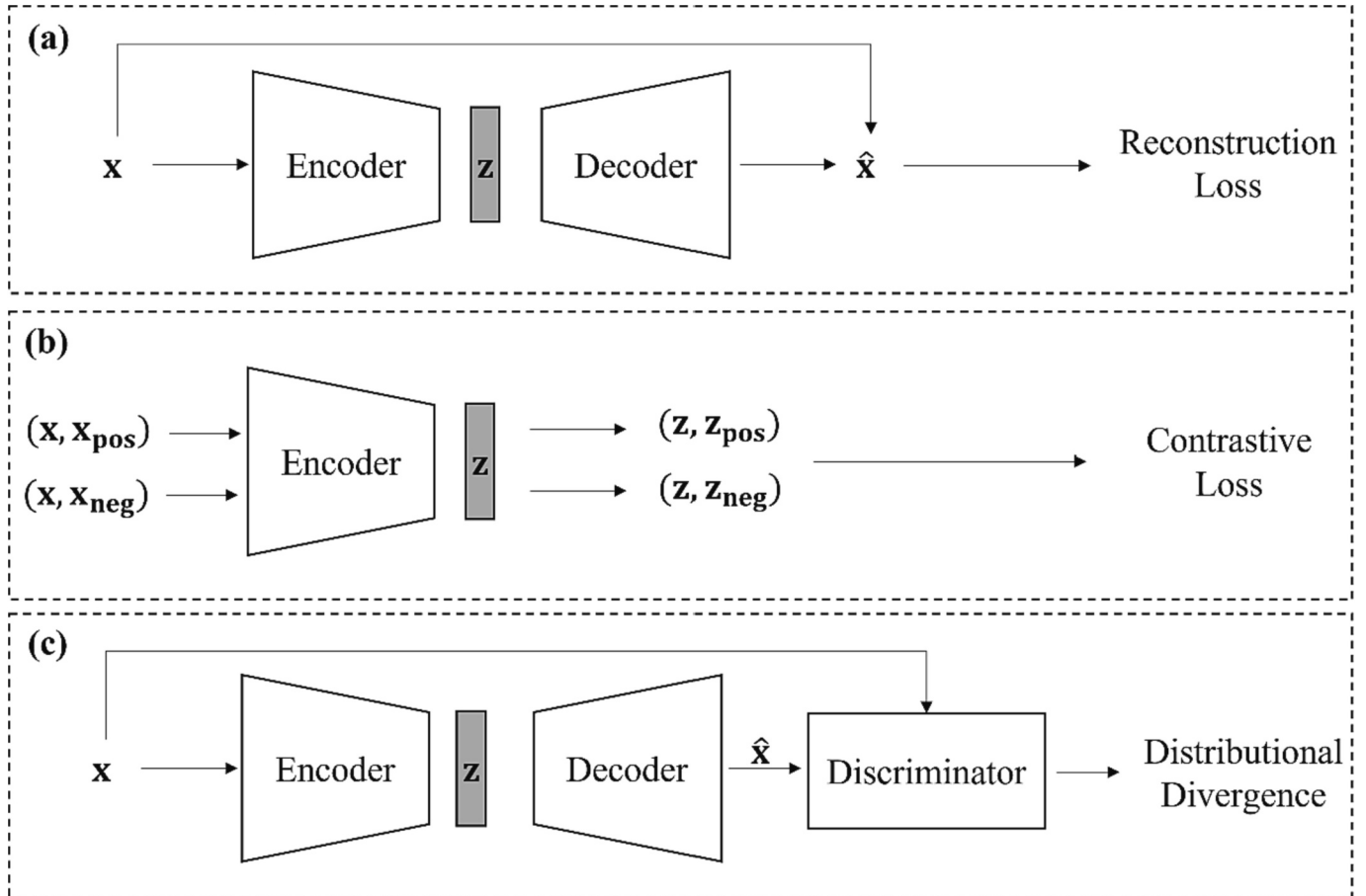


Fig. 9. The framework of (a) generative-based SSL, (b) contrastive-based SSL, (c) generative-contrastive SSL.

3.5. Self-supervised Learning

The aforementioned TL techniques can effectively reduce the reliance on labeled target samples but still require training samples from the source domain. However, it still takes time to identify a related source domain where we can learn relevant knowledge and label data from this domain. One promising way to reduce the reliance on source labels and achieve label efficiency is self-supervised learning (SSL), where a model learns to predict certain properties or features of input data without the need for explicit labels or annotations (Jaiswal et al., 2020; Liu et al., 2023b). SSL introduces supervisory signals by constructing pretext tasks that serve as a form of supervision.

According to the model architectures and learning objectives, the mainstream SSL methods can be summarized into three categories (Liu et al., 2023b), including generative-based SSL, contrastive-based SSL, and generative-contrastive SSL (Fig. 9). Specifically, generative-based SSL commonly employs the autoencoder architecture that learns to reconstruct the input data (Fig. 9 (a)). It consists of an encoder that compresses the input data x into a lower-dimensional representation z , and a decoder that outputs the reconstructed data $\hat{x}$. By minimizing the reconstruction error, it learns useful representations of the data. Examples include predicting the next word given previous words in natural language processing (NLP) (Brown et al., 2020). Contrastive-based SSL involves training an encoder to acquire representations z that capture shared characteristics among similar samples (positive pairs (x, x_{pos})) while discerning differences from dissimilar samples (negative pairs (x, x_{neg})) (Fig. 9 (b)). Typically, positive pairs consist of two diverse perspectives of the same sample, obtained through random data augmentation, while negative pairs are formed using unrelated samples from the dataset (He et al., 2020). By minimizing a contrastive loss, the encoder is trained to extract meaningful features z and distinguish relevant patterns within the unlabeled data, thereby facilitating improved generalization across downstream tasks. Examples include identifying the correct position of the scrambled patches in an image (Misra and Maaten, 2020). Generative-contrastive SSL, also known as adversarial-based SSL, utilizes the autoencoder architecture as well. An additional component called the discriminator is introduced and trained alongside the autoencoder (Fig. 9 (c)). The role of the discriminator is to differentiate between the generated data $\hat{x}$ produced by the autoencoder and the original input data x . The autoencoder and discriminator are trained in an adversarial manner to minimize the distributional divergences between the generated data and the original data. This interplay between the autoencoder and discriminator promotes the development of high-quality representations z and helps the model generalize well to downstream tasks. Examples include colorization (Cheng et al., 2015) and inpainting (Iizuka et al., 2017).

SSL methods are applicable when there are no labeled samples in the source domain. During the SSL phase, the model is trained to solve pretext tasks that require understanding and capturing relevant information from the data without explicit labels. This allows the model to leverage the vast amounts of unlabeled data available and learn valuable knowledge. The learned representations can be adapted for specific downstream tasks through techniques such as fine-tuning, where the SSL-trained model is further trained on a smaller labeled dataset specific to the downstream task.

In summary, the optimal TL technique for a particular task depends on various factors such as the availability of labeled data, the relatedness of the source and target learning tasks, and the goals of the learning tasks. Specifically, if a limited number of labeled data in the target domain is available, FTL can be considered to leverage knowledge from pre-trained models in a data-abundant source domain. If there are multiple related tasks at hand, MTL can be applied to improve the performance of one or multiple tasks by sharing information among them. If one is mainly dealing with prior shift and the objective is to generalize the model to new tasks, FSL would likely be the optimal choice. UDA is

most appropriate when the source and target domains have the same learning task, and no labels are available in the target domain. Finally, SSL is best used when there is no suitable source domain and large amounts of unlabeled data are available.

In addition, hybrid transfer learning (HTL), which combines two or more different TL approaches, can be an effective technique in certain conditions. For example, there is research that combines FTL and UDA by using pre-trained deep models as the backbone for UDA (Sicilia et al., 2023). Researchers also train models with unlabeled remote sensing imagery via SSL to learn representative features. The trained models are then adapted to downstream tasks via FTL (Cong et al., 2022). HTL can address some limitations of individual TL techniques. However, the advantages will depend on the specific application and the quality of the data used.

4. Applications

We conducted a systematic search for peer-reviewed articles, data papers, and reviews published between 2017 and 2022 in Web of Science. In the search process, we have identified key research topics and relevant terms for refinement. For example, when searching for literature on employing TL to map land cover using remote sensing data, we specify that the topic should encompass “Transfer Learning”, “Land Cover Mapping”, and “Remote Sensing”, while the publications should fall within the timeframe of “2017–2022”. The obtained results have undergone a meticulous screening procedure, considering factors such as titles, keywords, and abstracts. Notably, some papers related to radiative transfer models were initially included due to the presence of the keyword “Transfer,” despite not utilizing TL. After filtering out these irrelevant papers, the search identified 1676 papers related to TL in environmental remote sensing in 2017–2022. The number of related publications has steadily increased from 80 in 2017 to 621 in 2022 (Fig. 10 (a)), indicating the increasing popularity of TL in environmental remote sensing research.

In terms of methodology, the most dominant approach is FTL, which represents 67.1% of the research landscape (Fig. 10 (b)). UDA is the second most prevalent, accounting for 19.5% of the studies. SSL, FSL, and MTL represent smaller proportions of the research, at 6.6%, 3.1%, and 3.8%, respectively. We have also summarized these studies according to the specific research categories, including land cover mapping, vegetation monitoring, soil property estimation, crop yield prediction, biodiversity monitoring, water resources management, and natural disaster management (Fig. 10 (c)). Among them, the most common areas of study are land cover mapping and vegetation monitoring, followed by natural disaster management, water resource management, and biodiversity monitoring. We further explore the relationship between research categories and methodology (Fig. 11). FTL is predominant in most of the research categories. MTL is frequently used in soil property estimation since soil properties tend to be related and can be estimated with the same set of predictors. FSL is not widely applied in environmental remote sensing, only appearing in three research categories. In contrast, UDA and SSL are more popular, especially in areas where optical imagery is a major observation, such as land cover mapping and natural disaster management. Overall, TL has been applied to a wide range of applications, and the increasing number of publications suggests its potential for further advancements in the field of environmental remote sensing.

In the rest of this section, representative TL studies in the seven categories of research areas are reviewed. From land cover mapping to natural disaster management, these studies demonstrate the potential of TL to improve the accuracy, efficiency, and scalability of models in environmental remote sensing.

4.1. Land cover mapping

Land cover mapping is the process of identifying land cover types in a

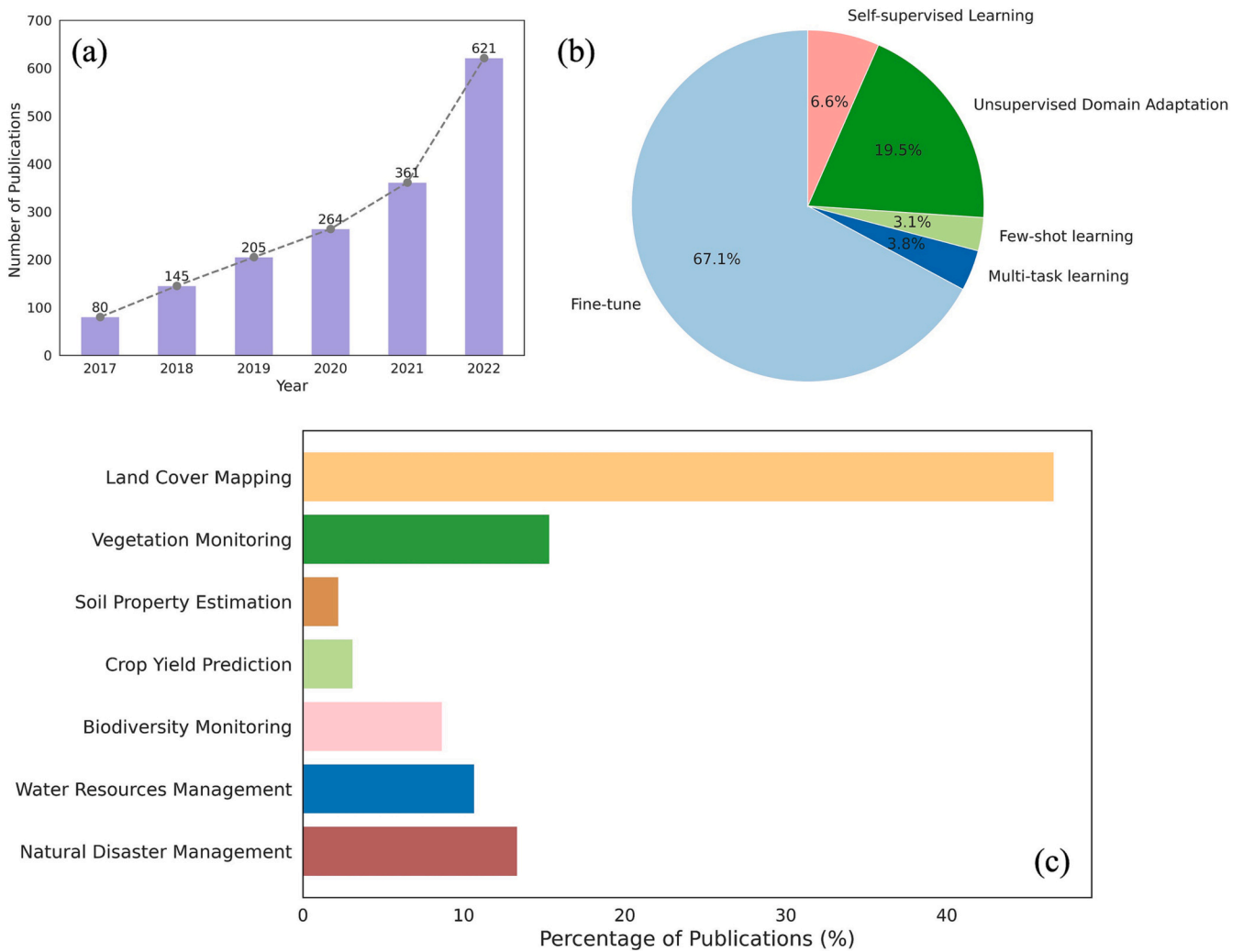


Fig. 10. The number and proportion of publications of TL in environmental remote sensing according to the (a) publication year, (b) methodology, and (c) research category.

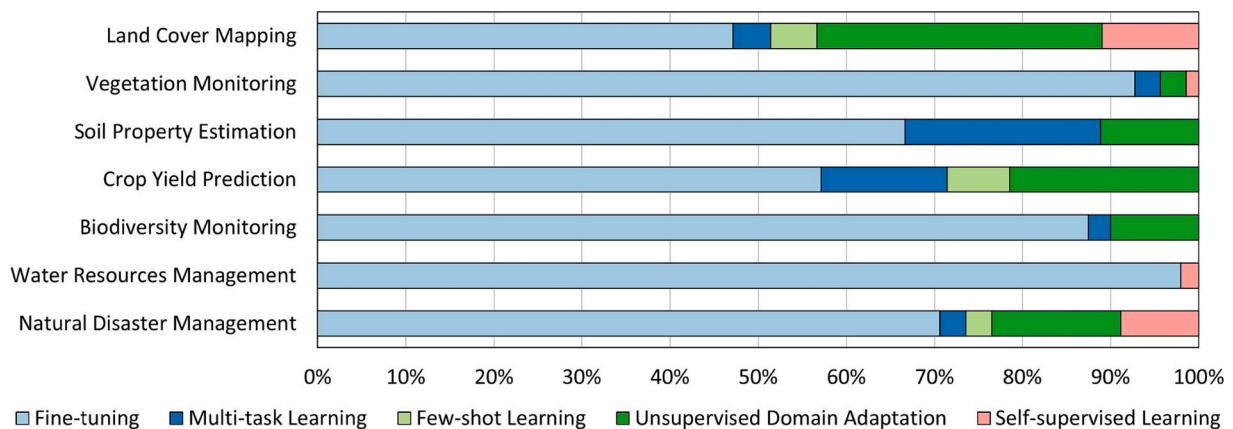


Fig. 11. The proportions of different TL methodologies used in each research category.

particular area. The composition and geospatial distribution of different types of land cover are critical for decision-making about environmental protection and agricultural management adaptation (Qian et al., 2020b). Land cover mapping commonly involves the use of remote sensing data, and the objective is to assign land cover labels to each pixel in remote sensing imagery (Marmanis et al., 2016; Panboonyuen et al.,

2019). Earth-observing satellites such as Landsat-8 and Sentinel-2 provide global coverage with adequate spatial and temporal resolutions for large-scale land cover mapping (Martini et al., 2021; Wang et al., 2021c). Researchers have also explored the potential of using UAVs for land cover mapping at the field level. These sensors provide the opportunity to collect a wealth of data but annotating these data requires

specialized skills and field surveys for validation. Land cover types and distributions also vary from place to place. As shown in Fig. 10, a clear difference can be observed in the Sentinel-2 remote sensing imagery and the corresponding CDL in five U.S. counties, indicating large domain shift between remote sensing observations over distant regions.

Researchers have applied TL to alleviate the need for annotation for land cover mapping in label-scarce regions. Some research focuses on general land-use and land-cover classification. For example, [de Lima and Marfurt \(2020\)](#) explored FTL for remote sensing scene classification using CNNs. They fine-tuned pre-trained models (Inception V3 and VGG19) to improve classification accuracy on three different remote sensing datasets (UC Merced, AID, and PatternNet). They found that fine-tuning well-established CNN models offers a practical way to achieve the best performance for remote sensing scene classification, with the best performance found when fine-tuning the intermediate layers of the models. Other studies focus on mapping specific types of crops ([Lin et al., 2022b; Nowakowski et al., 2021; Wang et al., 2019](#)). The CDL, which is produced annually by the United States Department of Agriculture (USDA) National Agricultural Statistics Service (NASS), provides crop-specific land cover in the U.S. at a 30-m spatial resolution (Fig. 12) ([Boryan et al., 2011](#)). It is commonly used as the ground truth label for crop mapping. Since high-quality cropland layers are not available in many other regions, TL studies on crop mapping commonly used the U.S. as the source domain and transfer it to other regions. For example, [Gadiraju and Vatsavai \(2020\)](#) collected aerial imagery from the National Agriculture Imagery Program (NAIP) and used CDL as the ground truth data. They compared the performance of four models, including a model trained from scratch in the target domain, the model transferred directly from the source domain, the FTL model from fine-tuning the bottom layers, and the FTL model from fine-tuning all layers. The FTL models by fine-tuning the bottom layers or all layers delivered the best results.

To achieve crop mapping at a high spatial resolution, some studies transfer models to map field-level crop types based on UAV imagery. For example, [Chew et al. \(2020\)](#) collected UAV imagery with 3 cm resolution and fine-tuned a pre-trained VGG16 architecture to fulfill the task of within-field crop mapping in Rwanda. Similarly, [Nowakowski et al. \(2021\)](#) fine-tuned pre-trained VGG16 and GoogLeNet models for crop mapping in Africa using local high-resolution RGB images collected by drone and field campaigns. The fine-tuned models outperformed the models trained from scratch and achieved nearly 83% accuracy in

Kasungu District, Malawi, and up to 90% in Gaza Province, Mozambique. They also evaluated the FTL performance given the number of frozen layers and found that fine-tuning all the CNN weights resulted in significantly better performance than the approaches that fine-tuned only some bottom layers.

UDA and SSL have also been used for crop mapping. For example, [Bejiga et al. \(2019\)](#) proposed a DANN model for vegetation mapping in Europe based on Landsat-8. [Martini et al. \(2021\)](#) further designed the feature extractor of DANN as a transformer encoder to handle time-series Sentinel-2 for crop mapping. The proposed model boosted the classification accuracy by 30%, associated with an evident reduction of the feature distance metrics. [Wang et al. \(2022a\)](#) extended the UDA framework for multi-source domain adaptation by designing domain-specific feature extractors and training the model to alternatively minimize source-target domain shift in time-series Sentinel 2 images. They treated eleven U.S. states as source domains and transferred the model for crop mapping in three provinces in China and achieved higher accuracy than non-transfer models and single-source UDA models (i.e., the UDA methods that assume that there is only one source domain), such as DANN.

Moreover, there are efforts to design SSL strategies for remote sensing model training. Based on the idea that neighboring images should be similar and distant images should be dissimilar, [Jean et al. \(2019\)](#) developed a contrastive-based SSL strategy for remote sensing imagery named Tile2Vec, in which the feature extractor is trained to minimize the Euclidean distance between the embeddings of the anchor tile and the neighbor tile while maximizing the distance between the anchor and distant embeddings. The learned Tile2Vec representations outperformed other unsupervised features on the CDL classification task in California. [Jung et al. \(2022\)](#) extended the Tile2Vec framework by simultaneously utilizing neighboring tiles as a positive pair of the anchor tile to minimize their contrastive loss and learn smoothed representation for the downstream task of crop mapping. However, the aforementioned SSL frameworks only considered spatial correlation in remote sensing imagery but ignored temporal dependency. More recent studies try to consider both spatial correlation and temporal dependency in time-series remote sensing observations. [Ayush et al. \(2021\)](#) proposed a geography-aware SSL strategy in which spatially aligned time-series satellite images are used to construct temporal positive pairs and geo-spatial information is used to design pre-text tasks. The proposed

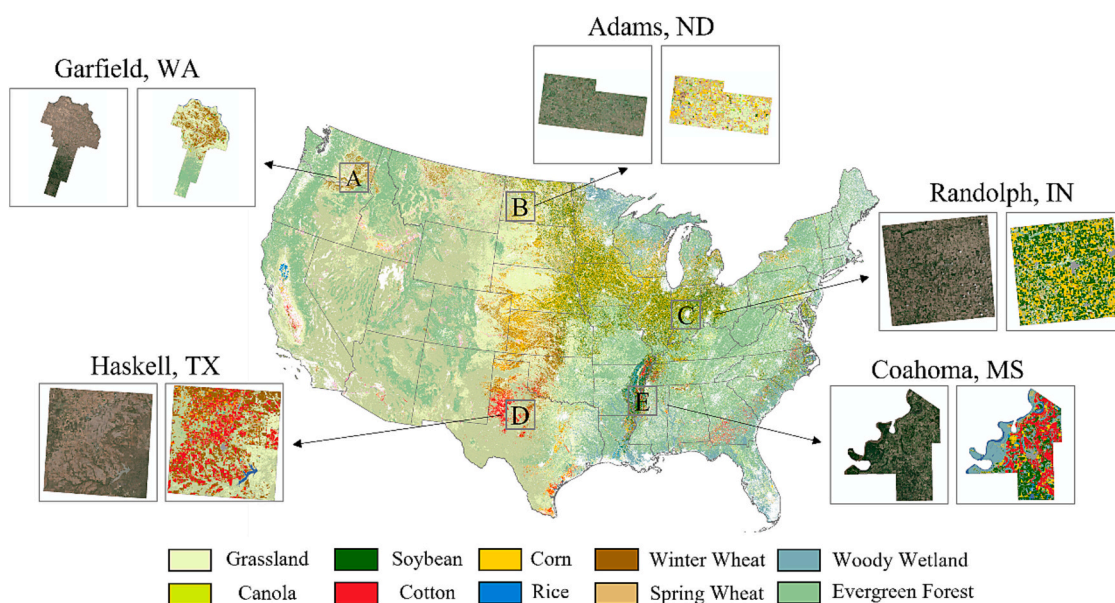


Fig. 12. Sentinel-2 remote sensing imagery and the corresponding cropland data layers (CDL) in five U.S. counties. Land cover types and distributions vary from place to place, resulting in domain shift between different spatial regions.

strategy was proven to improve the accuracy of landcover type classification and semantic segmentation. Inspired by Masked Autoencoder (MAE), Cong et al. (2022) further presented a SSL framework named SatMAE for temporal and multi-spectral satellite imagery. Both temporal encoding and spatial encoding are considered, and multi-spectral data is encoded with distinct spectral positional encodings in SatMAE. Up to 14% improvement was achieved by SatMAE in land cover classification compared to previous state-of-the-art techniques.

Overall, TL approaches have achieved promising results. However, previous studies have primarily focused on addressing spatial domain shift across geographic regions but have overlooked temporal domain shift that can arise due to variations in observation time. For example, since the planting progress varies each year, the corresponding time-series reflectance values in the same regions can be different from year to year, leading to temporal domain shifts in remote sensing observations. Also, a large portion of existing research on land cover mapping based on UAV imagery uses ImageNet as the source domain for model pre-training and then fine-tunes the model on local UAV imagery. However, the degree of relatedness between ImageNet and UAV imagery is uncertain. Further research may consider using satellite remote sensing imagery as the source domain and transfer the knowledge for high-resolution land cover mapping on UAV imagery.

4.2. Vegetation monitoring

Remote sensing is widely used to monitor vegetation dynamics, including vegetation health, biomass estimation, and vegetation productivity. Specifically, the physical and biological properties of vegetation are retrieved and quantified as vegetation parameters, such as Leaf Area Index (LAI) (Li et al., 2021b), biomass (Wang and Crawford, 2021), leaf chlorophyll content (Zhang et al., 2021), fractional vegetation cover (Yu et al., 2022), and leaf nitrogen concentration (Wan et al., 2022). These parameters are important for understanding the structure, function, and health of ecosystems. Traditional vegetation parameter retrieval methods involve using handheld instruments during field work, which are challenging to scale up and expensive in time and resources. In the case of measuring LAI, for example, the commonly used handheld LAI-2000 plant canopy analyzer needs to be positioned below the vegetation canopy and moved around the measurement site, with multiple measurements at each location (Solberg et al., 2006).

Given the difficulty of acquiring training data for ML and DL models, TL has been widely explored for vegetation parameter retrieval. For example, to achieve low-cost LAI estimation for winter wheat, Li et al. (2021b) acquired RGB images over wheat fields using a Canon 600D digital camera mounted on a tripod at a height of 1.5 m from the ground. They pre-trained CNN models with data in 2018 and effectively improved the transferability by fine-tuning them with data in 2019. Li et al. (2022) proposed a UDA method to reduce domain shift between the Moderate Resolution Imaging Spectroradiometer (MODIS) surface reflectance dataset and the Visible Infrared Imaging Radiometer Suite (VIIRS) surface reflectance dataset. The TCA algorithm was first utilized to map the MODIS and VIIRS datasets into the same subspace to reduce domain shift. Then, a general regression neural network was trained based on the embedded data obtained from MODIS along with the ground truth LAI values. Finally, the trained model was directly applied to the embedded data from the VIIRS dataset for retrieving the LAI values. Similarly, Wan et al. (2022) developed a UDA method by coupling TCA with the support vector regressor to transfer leaf nitrogen concentration assessment models across different plant species. Compared to the baseline model, the UDA model greatly improved the model's transferability by reducing the average error by 36.76%.

Instead of using data in certain years or certain regions for pre-training, some studies pre-train the DL model with simulated data generated by process-based models and fine-tune the model with a few in-situ field samples, namely process-guided machine learning (PGML) (Liu et al., 2022; Wang et al., 2023). It can be a cost-effective alternative

to collecting and labeling real-world data since simulated data can be generated quickly and can include a variety of environmental conditions that are not easily accessible in real-world data. For example, Zhang et al. (2021) pre-trained a DNN model for the winter wheat LCC with a simulated dataset generated by a radiative transfer model (RTM). The pre-trained model was fine-tuned with labeled samples in field observations to improve the model's accuracy in different field scenes. Similarly, Yu et al. (2022) generated a large number of simulated training samples with an RTM. An LSTM network was pre-trained with the simulated training dataset obtained and limited real samples from satellite images were used to fine-tune the pre-trained network to estimate fractional vegetation cover. Wang et al. (2023) leveraged domain knowledge by pre-training models using RTM-based synthetic simulations with various vegetation and soil conditions and fine-tuned the models with field measurements of cover crop traits and airborne hyperspectral reflectance. Compared to pure data-driven and RTM inversion approaches, this method reduced the root mean square error by nearly 15% in aboveground biomass estimation and 6% in canopy nitrogen content estimation.

FTL is the main approach used in existing TL studies on vegetation monitoring and vegetation parameter retrieval. Further research can explore the effectiveness of other TL techniques in this scope. For example, since some vegetation parameters are correlated, such as LAI and biomass, MTL can be applied to improve the model performance on retrieving every single parameter. Also, the acquisition of in situ vegetation samples and environmental variables of the long time series is still challenging in current studies. A new type of environmental remote sensing technique, namely Global Navigation Satellite System Reflectometry (GNSS-R), provides a new mode for acquiring in situ vegetation parameters. Therefore, a valuable research direction is to use TL approaches to fuse observations of multi-source data, including optical, microwave, and GNSS-R, to retrieve vegetation and environmental parameters (Yuan et al., 2020).

4.3. Soil property estimation

Soil property estimation refers to the process of determining the physical, chemical, or biological properties of soil, such as texture, nutrient content, organic carbon, soil moisture, pH value, or water-holding capacity. Traditional approaches of ground-based and in-situ measurements as well as laboratory analysis can provide accurate and precise measurements of soil properties but tend to be time-consuming and expensive, especially if large numbers of samples need to be analyzed (Li et al., 2021a). Efforts have been made over the last several decades to map soil properties at large scales with fine spatial and temporal resolutions where traditional measurements provide the most-accurate soil dataset which supports the calibration, validation, and evaluation of the large-scale digital soil mapping (McBratney et al., 2003; Minasny and McBratney, 2016). More recently, ML and DL have been increasingly employed to estimate soil properties from remote sensing observations and associated land surface properties, such as elevation, slope, and surface temperature. For example, ML models were developed to infer surface soil property and soil types based on satellite observations such as IKONOS imagery (Sullivan et al., 2005) or ground-penetration radar imagery (Odhiambo et al., 2004).

Relationships learned by ML models are commonly site-specific owing to variations in soil texture, surface roughness, and other physical components of soil. For example, while soil spectral libraries provide spectral data and corresponding soil properties across the globe, modeling of soil properties with such diverse and hyperdimensional soil spectral libraries do not generalize well locally due to domain shift (Shen et al., 2022; Wang et al., 2022b). To address it, Liu et al. (2018) proposed a one-dimensional convolutional neural network (1D-CNN) to retrieve soil clay content and used FTL to transfer knowledge from a large-scale soil spectral library to the local hyperspectral imagery dataset. They first pre-trained their 1D-CNN model using the LUCAS soil spectral library

(Tóth et al., 2013), which contains thousands of geo-referenced soil samples collected across Europe. The pre-trained model was then fine-tuned with airborne hyperspectral data and the corresponding soil clay content from southeastern Spain. The final model achieved a higher R^2 value of 0.76 compared to the pre-trained 1D-CNN model without fine-tuning, which only had an R^2 value of 0.38. Similarly, Shen et al. (2022) pre-trained a 1D-CNN model with three country-specific soil spectral libraries in China, Sweden, and the U.S. The model was fine-tuned with local soil spectroscopy to localize the soil organic carbon estimates in individual fields in each country and reduced the root mean square error by 25.8% compared to local modeling. Other research has demonstrated accuracy gains from MTL. For example, Ng et al. (2019) combined measurements from two spectrometers and used MTL to simultaneously estimate multiple soil properties, including sand, clay, total carbon content, organic carbon content, cation exchange capacity, and pH values, and achieved high R^2 of more than 0.95 for all target soil properties.

Although researchers have explored TL for soil property estimation, there are still relatively few studies in this area. The mainstream TL studies for soil property estimation based on remote sensing commonly use large soil spectral libraries as source domains and transfer both instances and representations to assist local modeling (Liu et al., 2018; Shen et al., 2022). However, soil spectral libraries are not updated frequently and cannot meet the needs of rapidly changing environments. For instance, soil properties such as soil organic matter are subject to change substantially after the land cover or land use changes (Hu et al., 2023). Also, most of these existing studies focus on the performance of TL models on a specific soil property within a certain region and do not investigate the generalizability of the models across different soil types and regions. Furthermore, current methods are mostly designed for bare soils. To improve the model's transferability across diverse environments, external factors, such as vegetation or crop residue coverage, could be considered in future TL studies of soil property estimation based on remote sensing.

4.4. Crop yield prediction

Accurate crop yield prediction is critical to food security since it provides timely information on the amount of food that will be available for consumption (Johnson, 2014). It can also help farmers plan for their

harvest, make informed decisions about planting and harvesting, and negotiate prices with buyers (Kang et al., 2020). In recent decades, ML-based crop yield prediction has made large progress and most of the current methods predict crop yield by establishing the association between crop yields and remote sensing vegetation indexes, often accounting for other factors such as weather, soil properties, irrigation, and fertilizer management (Kang et al., 2020; Sun et al., 2020). However, collecting yield records for model training involves time-consuming field works, labor-intensive destructive sampling, or a large-scale agricultural census. The lack of historical yield records for model training is a common constraint. Moreover, domain shift often occurs between agricultural regions because of different crop yield levels and unique environmental and climatic conditions, as illustrated in Fig. 13 for the U.S.

To address the gaps, TL has been increasingly used for crop yield prediction, especially on commodity crops (i.e., corn, soybean, wheat). For example, Wang et al. (2018b) trained an LSTM model in Argentina for soybean yield prediction based on MODIS spectral bands. The trained model was then adapted to Brazil via FTL, which outperformed the local model trained with Brazil data from scratch. Zhao et al. (2022a) simulated above-ground biomass and yield for winter wheat by using the Simple Algorithm For Yield model. The simulated data were used to pre-train a four-layer fully-connected network and the in-field measured data were used to fine-tune the model. There are also studies exploring MTL of simultaneous crop yield prediction for different crops or for the same crop at different time windows. Nguyen et al. (2019) present a spatial-temporal MTL algorithm to simultaneously predict within-field cotton yield in Texas in three years using environmental and field spectrum data. Khaki et al. (2021) adopted MTL for simultaneous corn and soybean yield prediction based on MODIS surface spectral reflectance in the U.S. corn belt at the county level. A CNN-based model called YieldNet was proposed to allocate the weights of the backbone feature extraction, which accurately predicted corn and soybean yield from one to four months before the harvest.

Although FTL and MTL have proven effective for crop yield prediction, some labeled samples from the target domain are still required. However, yield records may be completely missing in certain regions. To make accurate yield predictions for regions without historical yield records, Ma et al. (2021b) utilized UDA for corn yield prediction based on MODIS-derived vegetation indexes and proposed an adaptive DANN

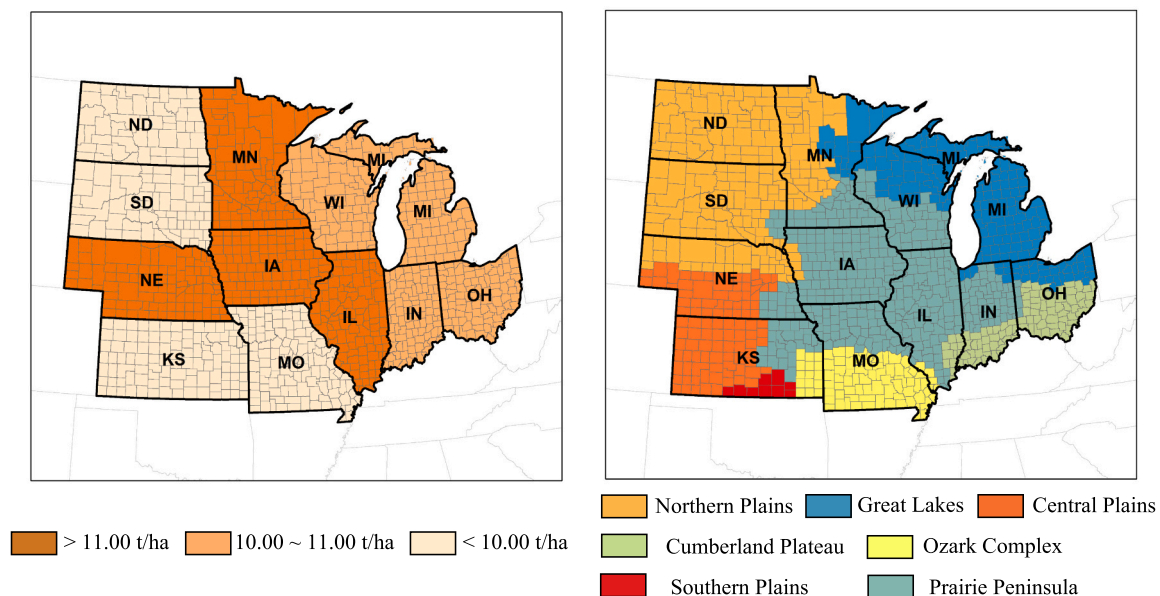


Fig. 13. Multiple domains in the U.S. corn belt based on state-level mean corn yield (Left) and National Ecological Observatory Network (NEON) eco-regions (Right) (Ma et al., 2023).

(ADANN) model to adaptatively adjust the weighting parameter between the yield prediction loss and the domain loss. Ma and Zhang (2022) further introduced Bayesian inference into ADANN and proposed Bayesian adversarial domain adaptation (BDANN) to improve the robustness of model training on a small training set. Both ADANN and BDANN have been evaluated in transfer experiments between two eco-regions in the U.S. corn belt and outperformed baseline models. Furthermore, to reduce the negative transfer caused by interference between dissimilar source samples in large-scale yield prediction, a multi-source maximum predictor discrepancy (MMPD) model was proposed by Ma et al. (2023) to selectively align the target domain with the most similar source domain by considering crop yield response via the domain-specific yield predictors in each domain (Fig. 14).

Few studies have explored TL for specialty crop yield prediction. For example, Chaudhary et al. (2021) used FTL for berries yield prediction based on MODIS land surface temperature and moisture products. They proposed a CNN-LSTM model with attention layers, which was pre-trained with station-based data and satellite images as inputs for the strawberry yield prediction and then transferred to predict the raspberry yield prediction via FTL. The performance obtained from FTL was comparable to that obtained by local modeling while having around a 55% reduction in processing time.

In summary, TL methods have proven useful for crop yield prediction. However, existing studies focused on regional yield prediction at the county level. To meet the need for precision agriculture, further research should concentrate more on field-level or subfield-level yield mapping. Also, current research concentrates on commodity crops, but few studies applied TL for specialty crop yield prediction, such as berries and nuts. Considering that specialty crops play an important part in local and regional food systems and contribute to the diversity and sustainability of agriculture, it is worth exploring the effectiveness of TL on yield prediction for specialty crops.

4.5. Biodiversity monitoring

The presence of a variety of species within an ecosystem is crucial for maintaining its proper functioning and ability to cope with environmental disruptions. Biodiversity plays a key role in determining both the

productivity and resilience of the ecosystem. (Cavender-Bares et al., 2020). The conventional method of field surveys, camera trapping, and visual inspection for monitoring biodiversity has limitations concerning the duration and area covered and often requires a significant investment of resources in terms of both time and money (Wang and Gamon, 2019). Recently, ML methods based on remote sensing have been applied for biodiversity monitoring, such as habitat mapping (Agrillo et al., 2021), species richness assessment (Choe et al., 2021), plant semantic segmentation (Schäfer et al., 2016), invasive species detection (Qian et al., 2020a). However, collecting labeled data for biodiversity monitoring still involves prolonged in-situ campaigns (Gogoll et al., 2020), which poses a limitation to the usage of ML and DL in biodiversity monitoring.

TL based on remote sensing has been increasingly adopted to monitor biodiversity, especially for estimating species composition and monitoring deforestation. For example, Zhu et al. (2020) present a FTL method to estimate phytoplankton species composition based on in situ hyperspectral remote sensing reflectance and hyperspectral satellite imagery (HICO images (Lucke et al., 2011)). Simulated data was used for pre-training and in situ observations were used to fine-tune the DNN model, which achieved high accuracy in predicting phytoplankton species composition in ocean regions. Soto Vega et al. (2021) used CycleGAN to change appearance and adapt Landsat-8 imagery across spatial domains. Additional constraints were added in the training phase to preserve the semantic structure and class transitions in the adapted images. Evaluation on the deforestation detection application in Amazon rainforest and Brazilian savanna using Landsat-8 images showed that the proposed CycleGAN effectively transferred the style of the target images that could accurately classified by the source classifiers. Soto et al. (2022) proposed a DANN + Change Vector Analysis framework to address unbalanced label distributions in the source domain and the target domain. A pseudo labeling scheme was devised to prevent the feature alignment from being biased toward the over-represented class in deforestation detection on Landsat-8 imagery.

In addition, TL, especially FTL, has also been used to detect invasive species in natural habitats or monitor weeds in agricultural fields. For example, Chaudhuri and Mishra (2023) employed TL to detect invasive purple loosestrife in the upper Mississippi river. A pre-trained ResNet-

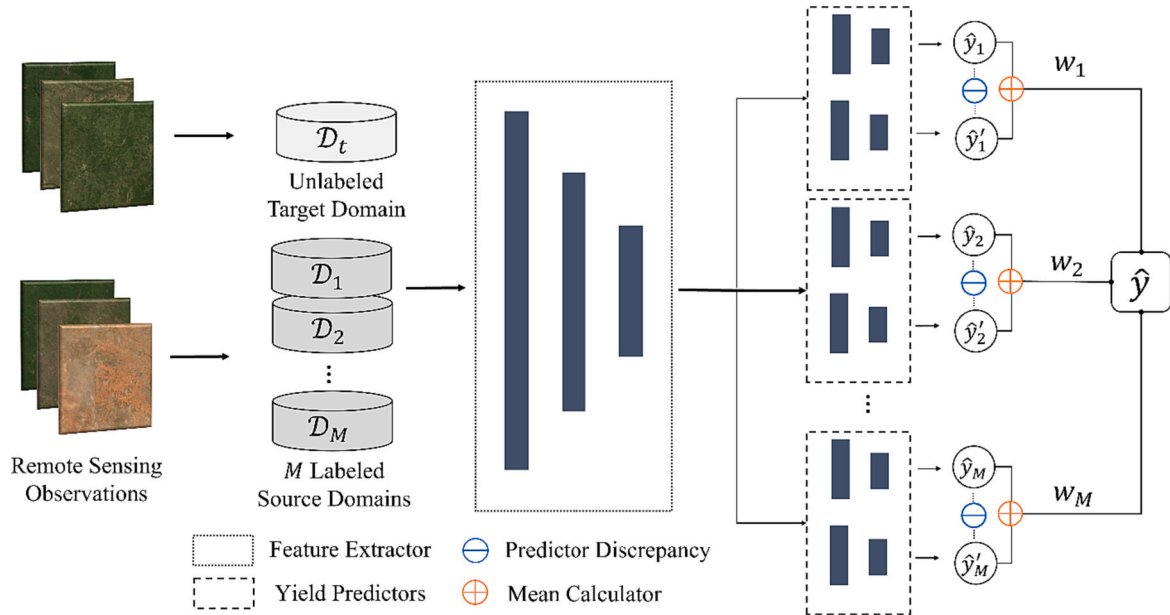


Fig. 14. The architecture of the multi-source maximum predictor discrepancy (MMPD) model with M source domains and one target domain (Reprinted from (Ma et al., 2023)). $\hat{y}_i$ and $\hat{y}'_i$ denote the predicted yields from a pair of yield predictors in the i th source domain and w_i is the corresponding weight. The final prediction $\hat{y}$ is the weighted average of predictions by predictors from all M source domains.

152 was used as the encoder and labeled local UAV imagery was used for model fine-tuning, which required less computation time and achieved high prediction accuracy when transferring to new study sites. Tufail et al. (2021) fine-tuned an Inception ResNet architecture for the recognition of *Ziziphus lotus* from aerial RGB images. Abdalla et al. (2019) used a VGG16-based encoder net for semantic segmentation of oilseed rapes from UAV images in a field with high-density weeds. The fine-tuned model was only used for feature extraction and the segmentation was performed using shallow ML classifiers, which achieved a high segmentation accuracy of 96% with fast inference time less than 0.05 s/image. Generative-based UDA methods have also been widely used to augment the training set by synthesizing data under different conditions via style transfer. For example, Zhao et al. (2021) used CycleGAN to synthesize field images in different seasons, under diverse illuminations, and in adverse weather conditions. They first trained a bale detection model using in-field UAV images captured with good illumination conditions in the fall and then trained the model with the synthesized images to improve the model's generalizability under different weather conditions. Bertoglio et al. (2023) presented a comparison of the two generative-based UDA for weed segmentation, including Fourier Transform and CycleGAN. They evaluated the models in two domain adaptation scenarios, one with images collected with different cameras and another with images collected by the same platform in different years. CycleGAN attained satisfying results in addressing domain shift caused by different cameras but both models struggled to address domain shift between different observation years.

In summary, recent years have witnessed increasing usage of TL in biodiversity monitoring based on remote sensing. Generative-based UDA has gained more adoption in semantic segmentation of plants and weeds compared to other environmental remote sensing applications. In this case, the generative models are trained to generate the crop/weeds images under different weather or illumination conditions and do not need to simulate complex geophysical responses for the tasks such as yield prediction, which makes the generative-based UDA more applicable. Despite the success, either FTL or UDA necessitates some time for model training and may not meet the requirements for real-time applications, which are essential for applications such as crop monitoring and robotic weeding (Chen et al., 2022b). Future TL research in the area can emphasize more efficiency and computational resources. In this respect, FSL is a promising strategy since it requires very few labeled target samples (i.e., few-shot learning) or there is even no need for labeled target samples (i.e., zero-shot learning) for model adaptation.

4.6. Water resources management

Water resources management is critical to environmental assessment and protection. For example, water availability assessment provides information about how much surface water is available and predictions of surface water quality provide early warnings for potential pollution incidents. Remote sensing together with ML techniques have been widely used for water resources management, such as estimating chlorophyll-a, nutrients, dissolved organic carbon, and water temperature (Cao et al., 2020; Harkort and Duan, 2023; Pyo et al., 2019). However, ground truth data collection is challenging for water resources management since in situ water sampling and lab analysis is required. Also, the quality and quantity of water can vary depending on the weather patterns and seasons. ML models trained using past water data can be challenging to handle new situations.

TL has been used to improve the temporal generalizability and spatial generalizability of ML models in water resources management. For example, Pu et al. (2019) chose Erhai Lake in Yunnan, China as the source domain and used in situ data and Landsat-8 imagery to train a CNN model for water quality classification. By FTL using a limited amount of in situ data, they adapted the pre-trained CNN to Chaohu Lake in Hefei, China, which is roughly 1.1 k miles away from the source domain. They proved that FTL was effective in increasing the spatial

generalizability of the CNN model and achieved high accuracy of water quality classification in the new area. To further reduce the use of in-situ data, Zhu et al. (2020) pre-trained a DNN with a semi-analytical model simulation dataset to predict phytoplankton species composition in the ocean. Hyperspectral remote sensing imagery and corresponding in situ samples were used to fine-tune the last few layers of the DNN model. Evaluation results in different ocean regions in China demonstrated considerable accuracy, highlighting the advantages of FTL in species composition estimation from hyperspectral remote sensing. Similarly, Miao et al. (2023) used ocean and atmosphere reanalysis datasets to pre-train a CNN model for long-term sea surface height anomalies (SSHAs) and sea surface temperature anomalies (SSTAs) prediction. Then, the pre-trained CNN model was fine-tuned by satellite-derived sea surface height, sea surface temperature, and wind speed data provided by Copernicus Marine and Environment Monitoring Service. The proposed FTL model successfully captured the evolving spatial patterns of SSHAs and SSTAs, displaying low prediction errors when considering a 30-day timeframe.

In addition, TL has also been used in studies of water availability, water consumption, and irrigation monitoring. For example, Wang et al. (2021a) transferred the optimal weights of a CNN model pre-trained on a Google dataset to the Landsat-8 dataset for water body extraction. The fine-tuned CNN was robust to the noisy and complex environment and achieved higher accuracy in extracting the water bodies in Tibetan Plateau. Dang and Li (2021) proposed a multiscale residual network based on SSL for water-body detection. The model outperformed the non-SSL model by a margin of ~1% in terms of overall accuracy in two water-body segmentation datasets of Gaofen satellite images with a spatial resolution of 0.5 m. To improve the training efficiency and model performance for irrigation system classification, Raei et al. (2022) selected a U-Net architecture with a Resnet-34 backbone pre-trained with ImageNet. After fine-tuning, the model achieved high accuracy in multiple irrigation-type segmentation on high-resolution aerial images with a spatial resolution of 1 m. Bazzi et al. (2020) concentrated on mapping irrigated areas based on time-series Sentinel-1 data. They transferred the model to address domain shift between geographic regions by first pre-training a CNN model on a source area with a huge volume of labeled samples and then refining the model by a few labeled samples from the target area. Similarly, Benbahria et al. (2021) used U-Net with a pre-trained ResNet50 backbone to map irrigated areas in Morocco from Landsat-8 images. They found that FTL with data augmentation significantly improved the mapping accuracy.

Although TL has been proven effective in water resource management, there are comparatively few studies in this field. Except for one study on SSL (Dang and Li, 2021), FTL appears to be the only TL technique used in water resource management, and there is a need to explore other TL strategies. Moreover, current studies focus on the optically active constituents but there are few works on optically inactive constituents, such as dissolved oxygen and cyanotoxins, since remote sensing alone cannot directly assess them (Ahmed et al., 2022). Future research can explore using multi-modalities and TL to extract and transfer knowledge from multiple data sources for accurate and systematic water quality monitoring.

4.7. Natural disaster management

Each year, natural disasters such as earthquakes, flooding, landslides, wildfires, and crop pest or disease outbreaks result in widespread loss of life, livelihood, and critical infrastructure globally. Remote sensing techniques together with ML algorithms have created valuable opportunities for disaster monitoring and management. High-resolution satellite platforms, airborne remote sensing platforms, and UAVs have provided efficient ways to track extreme natural events. ML and DL algorithms are used to interpret and analyze the remote sensing observations for applications such as building damage assessment (Dong and Shan, 2013; Nex et al., 2019), wildfire detection (Allison et al., 2016),

flood mapping (Peng et al., 2019), landslide detection (Mohan et al., 2021), plant disease identification (Too et al., 2019), and pest detection (Liu and Wang, 2021). Although powerful, it is challenging to obtain sufficient labeled data to train reliable models in newly disaster-stricken areas. Additionally, the process of identifying disaster pixels in remote sensing observations and annotating them can be a time-consuming task that requires specialized knowledge, which can impede rapid disaster response efforts.

TL has been used to transfer knowledge gained from historical disaster data to train models to facilitate faster and more efficient disaster response. Xu et al. (2020) fine-tuned a VGG-16 backbone for post-earthquake scene recognition with high-resolution Google Earth images of earthquake-stricken areas. The proposed FTL framework achieved higher overall recognition accuracy as well as improved the detection time by 95% compared to the traditional SVM method. Similarly, Lin et al. (2022a) applied FTL to address the discrepancy of regional building characteristics between historical and new earthquake-affected areas captured by remote sensing observations with different spatial resolutions and spectral bands. Instead of directly re-training the pre-trained model, they proposed a selective algorithm to retain the eligible source samples and remove those that differ significantly from the target data, which were used together with the labeled target samples to jointly fine-tune the model and achieved about 8% higher in overall accuracy for seismic building damage classification. Ahmad et al. (2019) proposed an FTL framework based on the Inception v3 architecture pre-trained on the ImageNet dataset to fuse satellite World-View3 images and geo-tagged social media images for the detection of passable roads after flooding. Both Park et al. (2021) and Khan et al. (2022) applied FTL to wildfire detection based on UAV optical images. They found that DL architectures such as VGG and ResNet pre-trained on ImageNet can be effectively adapted to detect local wildfires after fine-tuning and outperformed non-transfer methods. Defang et al. (2021) fine-tuned a pre-trained VGG-19 with 30 samples of optical remote sensing images from the GaoFen-2 satellite for landslide detection and achieved an accuracy of 98.22%, which is much higher than the accuracy of 64.68% by the direct learning model. Lu et al. (2020) also used the ImageNet dataset as the source domain and transferred useful information to boost the accuracy of landslide extraction in optical UAV images. The proposed CNN-based FTL framework has shown advantages for large-scale and scattered landslide extraction.

TL has also shown promise in the field of plant pest and disease management based on remote sensing. Some studies focused on classifying pests or diseases for a certain crop type. For example, Wang et al. (2017) compared several CNN-based architectures for the automated and accurate diagnosis of the severity of apple black rot disease via FTL. They found that the performance of fine-tuned models was superior to that of models trained from scratch and the best result was achieved by the VGG-16 model. Abbas et al. (2021) proposed a generative-based UDA framework to generate synthetic images of tomato plant leaves. A DenseNet121 model was fine-tuned on both synthetic and real images to classify the tomato leaves images into 10 categories of diseases. In addition, some studies developed TL models to classify diseases or insects in various field crops. Wang et al. (2018a) pre-trained CNN-based models with the PlantVillage dataset and fine-tuned them with in-field images to adapt the model for the classification of crop disease in the target area. Paymode and Malode (2022) fine-tuned the VGG model for disease detection of both tomatoes and grapes, which achieved a high accuracy of 98.40% of grapes and 95.71% of tomatoes.

For now, most studies focus on post-disaster management while a few of them are about disaster prevention and prediction. For example, Ai et al. (2022) pre-trained a fully-connected neural network to estimate landslide susceptibility by using a dataset of landslides in the Wenchuan earthquake and corresponding influencing factors (DEM, satellite-derived NDVI, weather variables). The model was then adapted to predict landslide susceptibility in a new region via FTL. Schmitter et al.

(2017) proposed a transformation-based UDA method to address domain shift in hyperspectral images for early detection of drought stress on barley. The method has been evaluated on datasets collected in different years or on different crops, and it improved the overall accuracy by 40% ~ 100% in comparison with non-UDA methods. To enable early warning of typhoon, Wang et al. (2021b) pretrained a Fast R-CNN on ImageNet and then fine-tuned it on imagery collected by China's geostationary meteorological satellite FY-2 to identify typhoon clouds and its center location on the ocean. The FTL method was proven to accurately detect typhoon clouds and outperformed non-TL methods by a margin of ~5% in terms of precision.

In summary, TL based on remote sensing has created valuable opportunities for reliable disaster monitoring and rapid disaster response. Current studies focus on post-disaster applications and management and more efforts should be put on natural disaster prediction and prevention. Also, FTL is the mainstream approach used in existing studies. There is a need to explore different TL strategies that can account for the dynamic and heterogeneous nature of disaster data. Moreover, existing research mostly uses ImageNet as the source domain which may not always provide useful knowledge for the tasks in the target domain. For example, Thenmozhi and Srinivasulu Reddy (2019) trained a deep CNN model from scratch for crop pest classification and compared its performance with pre-trained DL architectures on ImageNet. The proposed CNN model outperformed all the FTL models and achieved the highest classification accuracy in three insect datasets. Their results highlight the need for careful selection of the source domain in future research.

5. Discussion and recommendations for future work

5.1. Organizing benchmark datasets

To date, FTL is the most widely used TL approach in environmental remote sensing (Fig. 10). Image recognition datasets, such as ImageNet (Deng et al., 2009) and COCO (Lin et al., 2014), are widely used as the source domain for model pre-training (Table 3), yet these datasets are not designed for remote sensing tasks and many unrelated object categories are included, such as electronic devices and household items. These unrelated object categories will lower the relatedness between the source and target domains, leading to decreased transferability. Therefore, some researchers have reported decreased model performance when utilizing ImageNet as the source domain for remote sensing tasks (Thenmozhi and Srinivasulu Reddy, 2019).

Several well-organized remote sensing datasets exist, as summarized in Table 3. However, most of them are case-specific. For example, UC Merced Land Use Dataset is a collection of 2.1 k RGB aerial images and 21 classes of land use labels with a spatial resolution of 0.30 m (Yang and Newsam, 2010). DeepSat consists of 400 k RGBN satellite images including 6 types of landcover classes (Basu et al., 2015). DeepWeeds (Olsen et al., 2019) includes 17 k images and 8 classes of species for the application of plant classification and weed detection. These datasets are normally built for certain challenges or competitions. They are also commonly collected from a certain region, resulting in high similarity between samples of the same class (Cheng et al., 2017). As a result, the sizes and diversity of most remote sensing datasets are comparatively small and may not serve as a good source domain to train TL models for environmental remote sensing applications.

TL's progress in computer vision has been aided by well-organized benchmark datasets. For example, Office-31 consists of 4.1 k images in 31 classes from three distinct domains, including the Amazon website (i.e., Amazon domain), web camera (i.e., Webcam domain), and digital SLR camera (i.e., DSLR domain) (Koniusz et al., 2017). However, current TL studies in remote sensing are conducted case by case. Researchers commonly choose one spatial region as the source domain and transfer the model to the other region (Lin et al., 2022a; Ma and Zhang, 2021; Wang et al., 2022b) or use synthetic data generated by process-based models or generative models as the source domain and treat the real-

Table 3

Summary of publicly available datasets that are commonly used for transfer learning studies in remote sensing.

Name	Usage	Size	Bands	Resolution	Location	Reference
ImageNet	Image Classification	~1 M images from 1000 classes	RGB	N/A	Online	(Deng et al., 2009)
COCO	object detection, segmentation, key-point detection, and captioning	328 k images from ~80 categories	RGB	N/A	Online	(Lin et al., 2014)
PlantVillage	Plant disease detection and classification	54 k leaf images divided into 38 categories by species and disease	RGB	N/A	USA	(Hughes et al., 2015)
DeepWeeds	Weed Classification	17 k images for 8 categories of weed species	RGB	N/A	Australia	(Olsen et al., 2019)
Sugar Beets	plant classification	12 k images with sugar beet and 9 categories of weed species	RGB-D, LiDAR	N/A	Germany	(Chebrolu et al., 2017)
UC Merced Land Use Dataset	land use aerial dataset	2.1 k images for 21 land use classes	RGB	0.3 m	USA	(Yang and Newsam, 2010)
DeepSat	land use satellite dataset	400 k images for 6 landcover classes	RGB + NIR	1 m	USA	(Basu et al., 2015)
AID	land use aerial dataset	10 k images for 30 aerial scene types	RGB	8 m/0.5 m	Global	(Xia et al., 2017)
SpaceNet	Building footprints and road networks satellite dataset	N/A	Multi-band and Multi-sensor	0.3–4 m	Global	(Van Etten et al., 2018)
Functional Map of the World	functional purpose of buildings and land use satellite dataset	~1 M images from 63 categories	4 bands or 8 bands	0.3–3 m	20 Countries	(Christie et al., 2018)
DOTA	object detection aerial dataset	11,268 images from 18 categories	RGB	0.65–1 m	Global	(Xia et al., 2018)
Satlas	land use satellite dataset	856 k tiles for 137 classes	RGB + NIR	10 m/1 m	Global/USA	(Bastani et al., 2022)
Agriculture-Vision dataset	field anomaly patterns segmentation	95 k aerial farmland images with 9 types of field anomalies	RGB + NIR	0.1 m	USA	(Chiu et al., 2020)
PASTIS	satellite dataset for semantic segmentation of agricultural parcels	2.5 k time series images with 124 k individual parcels and 18 crop types	10 spectral bands from Sentinel-2	10 m	French	(Garnot and Landrieu, 2021)
LUCAS soil spectral library	Soil Property Estimation	20 k soil spectroscopy	Soil spectra in the 400–2500 nm range	N/A	Europe	(Tóth et al., 2013)

world data as the target domain (Sapkota et al., 2022; Zhang et al., 2021). There are no widely recognized benchmark datasets to evaluate and compare different algorithms, which hinders the development and evaluation of TL algorithms in remote sensing.

In summary, a key need exists for benchmark datasets that are explicitly designed for remote sensing applications and that include a diversity of data across locations and years. The state-of-the-art very-large-scale benchmark dataset in remote sensing is Satlas, which comprises 856 k tiles of Sentinel-2 multispectral images and 46 k tiles of high-resolution aerial images from the NAIP under 137 categories (Bastani et al., 2022). Pre-training on Satlas improved the average accuracy on downstream remote sensing tasks like road segmentation and ship detection by 16% over pre-training on ImageNet (Bastani et al., 2022). Although promising, more evaluation of Satlas on different downstream tasks is needed to demonstrate its effectiveness and feasibility. Another way to promote TL studies in environmental remote sensing is to organize TL challenges. Some attempts include the WILDS dataset challenge, in which 10 benchmark datasets that represent distribution shift faced in the wild are provided and a leaderboard has been established to show test performance by researchers from the world (Koh et al., 2021). Three of the benchmark datasets in WILDS involve remote sensing data (i.e., GlobalWheat, FMoW, PovertyMap) and the performances of TL algorithms can be tracked on the leaderboard.

5.2. Scale transfer via transfer learning

Another promising direction for TL research is to generate remote sensing products with higher spatial resolution. Many existing satellite products are produced at fairly broad scales, such as global land cover type (Friedl et al., 2010), precipitation products (Hou et al., 2014), LAI products (Yan et al., 2018), land surface temperature products (Park et al., 2005). A common route to improving the spatial resolution of these products has been statistical downscaling (SD), which establishes an empirical relationship between large-scale variables and local/regional variables and then uses it to generate high-resolution (HR) outputs of selected variables from low-resolution (LR) products (Ducournau and Fablet, 2017; Park et al., 2022; Rodrigues et al., 2018). Despite the effectiveness, loss of information can happen when

generating HR information, especially in cases where the original LR products are already noisy or blurry. SD also commonly assumes the statistical relationships between large-scale and local variables remain unchanged over time, which can lead to biased downscaled projections.

An alternative approach is to realize scale transfer via TL. In the setting of scale transfer, LR remote sensing observations and HR remote sensing observations are used as the source domain and the target domain, respectively. The TL approaches can be used to reduce domain shift caused by differences in spectral ranges and sensor configurations. Subsequently, HR remote sensing data can be utilized to create HR products in line with the same relationship between the LR remote sensing data and their corresponding LR labels. This approach offers a high degree of flexibility since the source and target domains can be selected as needed. Additionally, since LR products are not required as input, scale transfer via TL can be applied to generate HR products in regions where there are no LR products available.

5.3. Improving model interpretability

The interpretability of TL models is essential for understanding how they work and for identifying the factors that influence their transferability and performance. With interpretable TL models, the input feature variables with high transferability can be identified, which will guide feature selection and model architectures. However, existing TL models in environmental remote sensing focus on training strategies and model performance but lack interpretability. These studies evaluated their model empirically but did not interpret the model performance. The lack of interpretability of TL models can limit their usefulness and adaptation. Therefore, improving the interpretability of TL models in remote sensing is crucial, particularly in the context of climate change, to enable better utilization of these models in tackling unforeseen issues.

One potential research direction is to quantify the contribution of each predictor to the TL model's transferability. By doing this, the most transferable features can be identified. For example, in the application of crop mapping based on satellite imagery, one can assess the transferability of different spectral bands. This can inform feature selection and engineering strategies to improve the performance of TL models. Another potential research direction is to interpret the causes of domain

shift. Remote sensing applications often involve multiple factors that can cause domain shift, but current studies typically address them without considering the specific type of domain shift. For example, in county-level crop yield prediction (Ma et al., 2023; Wang et al., 2018b), researchers collected remote sensing observations over multiple years for model training and transferred the model from one agricultural region to another. The primary aim was to tackle domain shift resulting from the spatial separation between the source and target regions. However, temporal shift in crop phenology may also occur due to data collection at different periods. In future research, additional investigation is required to comprehend the underlying factors contributing to domain shift, and corresponding tactics could be developed to tackle each type of domain shift appropriately.

5.4. Foundation models for remote sensing

Foundation models (FMs), also referred to as large pre-trained models, are trained on large-scale data in a task-agnostic fashion and can be adjusted to a variety of downstream tasks using TL (Lu et al., 2023; Mai et al., 2023; Zhao et al., 2023). Such FMs allow for the transfer and sharing of knowledge across domains and mitigate the need for task-specific training data. Examples of FMs include (1) large language models such as LLAMA (Touvron et al., 2023) and ChatGPT (Liu et al., 2023a); (2) large vision models such as DALL-E2 (Ramesh et al., 2022) and Segment Anything Model (SAM) (Kirillov et al., 2023); (3) large multimodal models such as CLIP (Radford et al., 2021) and GPT-4 (OpenAI, 2023); and (4) large reinforcement learning models like Gato (Reed et al., 2022).

These FMs are generally trained on very large datasets and have demonstrated remarkable capabilities in downstream tasks in NLP and computer vision (CV). For example, SAM was trained with over 1 billion masks on 11 M images, which was the largest segmentation dataset by its release in April 2023. With a transformer-based image encoder and a prompt encoder, SAM performs impressively on image segmentation and can be easily adapted to downstream tasks, such as medical image analysis (Mazurowski et al., 2023) and civil infrastructure defect assessment (Ahmadi et al., 2023). There are also ongoing discussions on the potential use of FMs in numerous fields, such as public health (Biswas, 2023a), education (Latif et al., 2023), climate change (Biswas, 2023b), agriculture (Lu et al., 2023), geospatial artificial intelligence (Mai et al., 2023). We think FMs will also play an important role in environmental remote sensing, especially for TL studies. First, FMs are commonly trained with billions of unlabeled data via SSL (Devlin et al., 2019). As such, FMs can extract robust and generalizable features and can be easily adapted to downstream tasks in environmental remote sensing via few-shot or zero-shot learning. Second, in traditional TL settings, we need to manually determine the source domain, which provides auxiliary knowledge that is limited to a narrow and specific scenario. With FMs, there is no need to designate a specific source domain since they already possess extensive knowledge from a wide range of domains. Finally, large multimodal models such as GPT-4 may promote new TL strategies in environmental remote sensing. Specifically, existing TL studies in remote sensing require the same kind of observations in each domain. With large multimodal models, we may be able to transfer knowledge among different data types, such as visual or textual prompt-based knowledge transfer. For example, USDA publishes weekly crop progress reports about planting, fruiting, and harvesting progress and the overall condition of selected crops in major producing states. Such textual data provide critical prior information and can be potentially used together with remote sensing observations for applications such as land cover mapping, vegetation monitoring, crop yield prediction. Moreover, FMs have proved to have the ability of in-context learning, enabling them to make inferences on new tasks by leveraging in-context examples (prompt) without changing the model parameters (Brown et al., 2020). It offers the advantage of not requiring model fine-tuning, but its implementation can be tricky due to the model's sensitive

responsiveness to in-context examples (Zhang et al., 2023). It is worth exploring in-context learning of FMs in environmental remote sensing, particularly concerning tasks like retrieving and selecting relevant in-context examples.

In addition to using state-of-the-art FMs for remote sensing applications, there is also a need to develop remote sensing-based FMs to meet the unique challenges of remote sensing tasks. Mai et al. (2023) have compared the performance of FTL CNN models (e.g., ResNet) with zero-shot visual-language FMs (e.g., OpenCLIP) on the task of aerial scene classification on the AID dataset (Xia et al., 2017). They observed a performance gap between zero-shot FMs models and FTL CNN models. To fill this gap, there is a need to develop remote sensing-based FMs by considering the uniqueness of remote sensing images and tasks, such as being task-agnostic, sensor-agnostic, spatiotemporally aware, and environmentally invariant (Mai et al., 2023). There are some initial attempts to build remote sensing FMs, such as SatMAE (Cong et al., 2022). More recently, in cooperation with IBM Research, NASA has released its first open-source FM for earth observation, named HLS Geospatial FM (Jakubik et al., 2023), for applications such as landcover classification, natural disaster detection, and yield prediction. However, existing remote sensing FMs are trained with single-source data and designed for certain downstream tasks. Continued endeavors are needed to improve the capability and generalizability of remote sensing FMs.

Considering that the transformer architecture is the fundamental component of most existing FMs, future works on remote sensing-based FMs will likely rely on transformer-based architectures. However, the attention mechanism in transformers involves pairwise interactions between all tokens in the input sequence, resulting in a quadratic time complexity. As a result, it generally requires longer training time compared to CNNs that only have linear time complexity (Keles et al., 2023). Also, positional embeddings are incorporated into transformers to provide the model with information regarding the token positions within the sequence. This approach mitigates the inductive bias present in CNNs but introduces more trainable parameters and bigger memory footprint, necessitating a larger amount of training data (Vaswani et al., 2017). Consequently, transformers commonly require larger datasets and increased computational resources than CNNs to achieve better performance. In addition to organizing large datasets (Section 5.1), there are two promising research directions in training transformer-based FMs for remote sensing. One is to design better tokenization and encoding strategies for remote sensing data. Transformers require tokenization, and it's not entirely obvious what's the best way to do it with remote sensing data. Also, unlike natural images, remote sensing images have stronger temporal and spatial dependencies as well as spatial hierarchies. There is also a recent study arguing that temporal-then-spatial encoding is more intuitive for time-series satellite imagery processing (Tarasiou et al., 2023). Further studies should consider those characteristics of remote sensing data and design the encoding strategies accordingly. The second is to design better training methods and improve training and fine-tuning efficiency. We have seen increasing efforts to accelerate the training of large models in NLP and CV communities. For example, Low-Rank Adaptation of Large Language Models (LoRA) has been proposed to quicken the fine-tuning of large language models by only updating trainable rank decomposition matrices in each layer (Hu et al., 2021). It reduces the number of trainable parameters by ten thousand times when fine-tuning GPT-3 for downstream tasks. Similar studies for remote sensing data would have great potential in adapting existing FMs for remote sensing applications.

6. Conclusion

In this paper, a systematic review has been presented on TL in environmental remote sensing from both the perspectives of methodology and application. Five commonly used TL approaches have been introduced and representative TL studies in seven categories of applications in environmental remote sensing have been reviewed. The

review shows that TL techniques have demonstrated great value in environmental remote sensing. Looking forward, we highlight the potential benefits of organizing large and diverse benchmark datasets for model evaluation, the promise of TL for scale transfer, the need to improve model interpretability, and the opportunity to leverage FMs for remote sensing tasks.

CRedit authorship contribution statement

Yuchi Ma: Conceptualization, Formal analysis, Writing – Original draft, Data curation, Writing – review & editing. Shuo Chen: Conceptualization, Investigation, Formal analysis, Writing – Original draft. Stefano Ermon: Writing – review & editing. David B. Lobell: Conceptualization, Funding acquisition, Resources, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Acknowledgements

We thank three anonymous reviewers provided helpful comments on the manuscript. Funding was provided by the NASA Acres (NASA Applied Sciences Grant No. 80NSSC23M0034, sub-award 124245-Z6512205 to D.B.L.). Any opinions, findings, and conclusions or recommendations expressed in this study are those of the authors and do not necessarily reflect the reviews of NASA.

References

- Abbas, A., Jain, S., Gour, M., Vankudothu, S., 2021. Tomato plant disease detection using transfer learning with C-GAN synthetic images. *Comput. Electron. Agric.* 187 <https://doi.org/10.1016/j.compag.2021.106279>.
- Abdalla, A., Cen, H., Wan, L., Rashid, R., Weng, H., Zhou, W., He, Y., 2019. Fine-tuning convolutional neural network with transfer learning for semantic segmentation of ground-level oilseed rape images in a field with high weed pressure. *Comput. Electron. Agric.* 167, 105091 <https://doi.org/10.1016/j.compag.2019.105091>.
- Agrillo, E., Filippini, F., Pezzarossa, A., Casella, L., Smiraglia, D., Orasi, A., Attorre, F., Taramelli, A., 2021. Earth observation and biodiversity big data for forest habitat types classification and mapping. *Remote Sens.* 13 <https://doi.org/10.3390/rs13071231>.
- Ahmad, K., Pogorelov, K., Riegler, M., Ostroukhova, O., Halvorsen, P., Conci, N., Dahyot, R., 2019. Automatic detection of passable roads after floods in remote sensed and social media data. *Signal Process. Image Commun.* 74, 110–118. <https://doi.org/10.1016/j.image.2019.02.002>.
- Ahmadi, M., Lonbar, A.G., Sharifi, A., Beris, A.T., Nouri, M., Javidi, A.S., 2023. Application of segment anything model for civil infrastructure defect assessment. *arXiv Prepr arXiv2304.12600*.
- Ahmed, M., Mumtaz, R., Anwar, Z., Shaukat, A., Arif, O., Shafait, F., 2022. A multi-step approach for optically active and inactive water quality parameter estimation using deep learning and remote sensing. *Water* 14, 2112.
- Ai, X., Sun, B., Chen, X., 2022. Construction of small sample seismic landslide susceptibility evaluation model based on transfer learning: a case study of Jiuzhaiguo earthquake. *Bull. Eng. Geol. Environ.* 81, 116.
- Allison, R.S., Johnston, J.M., Craig, G., Jennings, S., 2016. Airborne optical and thermal remote sensing for wildfire detection and monitoring. *Sensors (Switzerland)* 16. <https://doi.org/10.3390/s16081310>.
- Ayush, K., Uzket, B., Meng, C., Tanmay, K., Burke, M., Ermon, S., Lobell, D., Ermon, S., 2021. Geography-aware self-supervised learning. *Proc. IEEE/CVF Int. Conf. Comput. Vis.* 10181–10190. <https://doi.org/10.1109/iccv48922.2021.01002>.
- Bastani, F., Wolters, P., Gupta, R., Ferdinando, J., Kembhavi, A., 2022. Satlas: a large-scale, multi-task dataset for remote sensing image understanding. *arXiv Prepr arXiv2211.15660*.
- Basu, S., Ganguly, S., Mukhopadhyay, S., DiBianio, R., Karki, M., Nemani, R., 2015. DeepSat - A learning framework for satellite imagery. In: *GIS Proc. ACM Int. Symp. Adv. Geogr. Inf. Syst.* 03–06-Nov. <https://doi.org/10.1145/2820783.2820816>.
- Bazzi, H., Ienco, D., Baghdadi, N., Zribi, M., Demarez, V., 2020. Distilling before refine: Spatio-temporal transfer learning for mapping irrigated areas using Sentinel-1 time series. *IEEE Geosci. Remote Sens. Lett.* 17, 1909–1913. <https://doi.org/10.1109/LGRS.2019.2960625>.
- Bejiga, M.B., Melgani, F., Beraldi, P., 2019. Domain adversarial neural networks for large-scale land cover classification. *Remote Sens.* 11, 1–20.
- Bellochio, E., Costante, G., Cascianelli, S., Fravolini, M.L., Valigi, P., 2020. Combining domain adaptation and spatial consistency for unseen fruits counting: a quasi-supervised approach. *IEEE Robot. Autom. Lett.* 5, 1079–1086. <https://doi.org/10.1109/LRA.2020.2966398>.
- Benbahria, Z., Sebari, I., Hajji, H., Smiej, M.F., 2021. Intelligent mapping of irrigated areas from Landsat 8 images using transfer learning. *Int. J. Eng. Geosci.* 6, 40–50. <https://doi.org/10.26833/ijeg.681312>.
- Bertoglio, R., Mazzucchi, A., Catalano, N., Matteucci, M., 2023. A comparative study of Fourier transform and CycleGAN as domain adaptation techniques for weed segmentation. *Smart Agric. Technol.* 4, 100188 <https://doi.org/10.1016/j.atech.2023.100188>.
- Biswas, S.S., 2023a. Role of chat GPT in public health. *Ann. Biomed. Eng.* 51, 868–869. <https://doi.org/10.1007/s10439-023-03172-7>.
- Biswas, S.S., 2023b. Potential use of chat GPT in global warming. *Ann. Biomed. Eng.* 3–4 <https://doi.org/10.1007/s10439-023-03171-8>.
- Boryan, C., Yang, Z., Mueller, R., Craig, M., 2011. Monitoring US agriculture: the US department of agriculture, national agricultural statistics service, cropland data layer program. *Geocarto Int.* 26, 341–358.
- Bouchard, I., Rancourt, M.E., Aloise, D., Kalaitzis, F., 2022. On transfer learning for building damage assessment from satellite imagery in emergency contexts. *Remote Sens.* 14, 1–29. <https://doi.org/10.3390/rs14112532>.
- Brown, T.B., Mann, B., Ryder, N., Subbiah, M., Kaplan, J.D., Dhariwal, P., Neelakantan, A., Shyam, P., Sastry, G., Askell, A., Agarwal, S., Herbert-Voss, A., Krueger, G., Henighan, T., Child, R., Ramesh, A., Ziegler, D.M., Wu, J., Winter, C., Hesse, C., Chen, M., Sigler, E., Litwin, M., Gray, S., Chess, B., Clark, J., Berner, C., McCandlish, S., Radford, A., Sutskever, I., Amodei, D., 2020. Language models are few-shot learners. *Adv. Neural Inf. Process. Syst.* 33, 1877–1901.
- Cao, Z., Ma, R., Duan, H., Pahlevan, N., Melack, J., Shen, M., Xue, K., 2020. A machine learning approach to estimate chlorophyll-a from Landsat-8 measurements in inland lakes. *Remote Sens. Environ.* 248, 111974 <https://doi.org/10.1016/j.rse.2020.111974>.
- Cavender-Bares, J., Gamon, J.A., Townsend, P.A., 2020. Remote sensing of plant biodiversity. *Remote Sens. Plant Biodivers.* <https://doi.org/10.3390/rs14112532>.
- Chadha, A., Andreopoulos, Y., 2019. Improved techniques for adversarial discriminative domain adaptation. *IEEE Trans. Image Process.* 29, 2622–2637.
- Chaudhary, M., Gastli, M.S., Nassar, L., Karray, F., 2021. Transfer Learning Application for Berries Yield Forecasting using Deep Learning. In: *Proceedings of the International Joint Conference on Neural Networks. Institute of Electrical and Electronics Engineers Inc.* <https://doi.org/10.1109/IJCNN52387.2021.9533530>.
- Chaudhuri, G., Mishra, N.B., 2023. Detection of aquatic invasive plants in wetlands of the upper Mississippi River from UAV imagery using transfer learning. *Remote Sens.* 15, 1–19. <https://doi.org/10.3390/rs15030734>.
- Chebrolu, N., Lottes, P., Schaefer, A., Winterhalter, W., Burgard, W., Stachniss, C., 2017. Agricultural robot dataset for plant classification, localization and mapping on sugar beet fields. *Int. J. Robot. Res.* 36, 1045–1052. <https://doi.org/10.1177/0278364917720510>.
- Chen, D., Lu, Y., Li, Z., Young, S., 2022b. Performance evaluation of deep transfer learning on multi-class identification of common weed species in cotton production systems. *Comput. Electron. Agric.* 198 <https://doi.org/10.1016/j.compag.2022.107091>.
- Chen, S., Liu, W., Feng, P., Ye, T., Ma, Y., Zhang, Z., 2022a. Improving spatial disaggregation of crop yield by incorporating machine learning with multisource data: a case study of Chinese maize yield. *Remote Sens.* 14, 2340. <https://doi.org/10.3390/rs14102340>.
- Cheng, G., Han, J., Lu, X., 2017. Remote sensing image scene classification: benchmark and state of the art. *Proc. IEEE* 105, 1865–1883. <https://doi.org/10.1109/JPROC.2017.2675998>.
- Cheng, Z., Yang, Q., Sheng, B., 2015. Colorful image colorization. In: *Proc. IEEE Int. Conf. Comput. Vis.* 2015 Inter, pp. 415–423. <https://doi.org/10.1109/ICCV.2015.55>.
- Chew, R., Rineer, J., Beach, R., O'neil, M., Ujeneza, N., Lapidus, D., Miano, T., Hegarty-Craver, M., Polly, J., Temple, D.S., 2020. Deep neural networks and transfer learning for food crop identification in UAV images. *Drones* 4, 1–14. <https://doi.org/10.3390/drones4010007>.
- Chiu, M.T., Xu, X., Wei, Y., Huang, Z., Schwing, A., Brunner, R., Khachatryan, H., Karapetyan, H., Dozier, I., Rose, G., Wilson, D., Tudor, A., Hovakimyan, N., Huang, T.S., Shi, H., 2020. Agriculture-vision: a large aerial image database for agricultural pattern analysis. *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit.* 2825–2835. <https://doi.org/10.1109/CVPR42600.2020.00290>.
- Choe, H., Chi, J., Thorne, J.H., 2021. Mapping potential plant species richness over large areas with deep learning, MODIS, and species distribution models. *Remote Sens.* 13, 1–20. <https://doi.org/10.3390/rs13132490>.
- Christie, G., Fendley, N., Wilson, J., Mukherjee, R., 2018. Functional map of the world. *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit.* 2, 6172–6180. <https://doi.org/10.1109/CVPR.2018.00646>.
- Cong, Y., Khanna, S., Meng, C., Liu, P., Rozi, E., He, Y., Burke, M., Lobell, D.B., Ermon, S., 2022. Satmae: pre-training transformers for temporal and multi-spectral satellite imagery. *Adv. Neural Inf. Process. Syst.* 35, 197–211.
- Dang, B., Li, Y., 2021. Msresnet: multiscale residual network via self-supervised learning for water-body detection in remote sensing imagery. *Remote Sens.* 13 <https://doi.org/10.3390/rs13161122>.

- Defang, L., Li, J., Fan, F., 2021. Classification of landslides on the southeastern Tibet plateau based on transfer learning and limited labelled datasets. *Remote Sens. Lett.* 12, 286–295.
- Deng, J., Dong, W., Socher, R., Li, L.-J., Li, K., Fei-Fei, L., 2009. ImageNet: A Large-Scale Hierarchical Image Database 248–255. <https://doi.org/10.1109/cvprw.2009.5206848>.
- Devlin, J., Chang, M.W., Lee, K., Toutanova, K., 2019. BERT: Pre-training of deep bidirectional transformers for language understanding. *NAACL HLT 2019–2019 Conf. North Am. Chapter Assoc. Comput. Linguist. Hum. Lang. Technol. - Proc. Conf.* 1, 4171–4186.
- Di Tommaso, S., Wang, S., Vajipey, V., Gorelick, N., Strey, R., Lobell, D.B., 2023. Annual field-scale maps of tall and short crops at the global scale using GEDI and Sentinel-2. *Remote Sens.* 15, 1–24. <https://doi.org/10.3390/rs15174123>.
- Dong, L., Shan, J., 2013. A comprehensive review of earthquake-induced building damage detection with remote sensing techniques. *ISPRS J. Photogramm. Remote Sens.* 84, 85–99. <https://doi.org/10.1016/j.isprsjprs.2013.06.011>.
- Ducournau, A., Fablet, R., 2017. Deep learning for ocean remote sensing: an application of convolutional neural networks for super-resolution on satellite-derived SST data. 2016 9th IAPR work. *Pattern Recognit. Remote Sensing, PRRS 2016*. <https://doi.org/10.1109/PRRS.2016.7867019>.
- Feng, L., Zhang, Z., Ma, Y., Sun, Y., Du, Q., Williams, P., Drewry, J., Luck, B., 2021. Multitask learning of alfalfa nutritive value from UAV-based hyperspectral images. *IEEE Geosci. Remote Sens. Lett.* 19, 1–5.
- Friedl, M.A., Sulla-Menashe, D., Tan, B., Schneider, A., Ramankutty, N., Sibley, A., Huang, X., 2010. MODIS collection 5 global land cover: algorithm refinements and characterization of new datasets. *Remote Sens. Environ.* 114, 168–182.
- Gadiraju, K.K., Vatsavai, R.R., 2020. Comparative analysis of deep transfer learning performance on crop classification. In: *Proceedings of the 9th ACM SIGSPATIAL International Workshop on Analytics for Big Geospatial Data, BIGSPATIAL 2020*. Association for Computing Machinery, Inc. <https://doi.org/10.1145/3423336.3431369>.
- Ganin, Y., Ustinova, E., Ajakan, H., Germain, P., Larochelle, H., Laviolette, F., Marchand, M., Lempitsky, V., 2017. Domain-adversarial training of neural networks. *Adv. Comput. Vis. Pattern Recognit.* 17, 189–209. https://doi.org/10.1007/978-3-319-58347-1_10.
- Garnot, V.S.F., Landrieu, L., 2021. Panoptic segmentation of satellite image time series with convolutional temporal attention networks. *Proc. IEEE Int. Conf. Comput. Vis.* 4852–4861 <https://doi.org/10.1109/ICCV48922.2021.00483>.
- Gogoll, D., Lottes, P., Weyler, J., Petrinic, N., Stachniss, C., 2020. Unsupervised domain adaptation for transferring plant classification systems to new field environments, crops, and robots. *IEEE Int. Conf. Intell. Robot. Syst.* 2636–2642 <https://doi.org/10.1109/IROS45743.2020.9341277>.
- Gu, X., Yu, X., Yang, Y., Sun, J., Xu, Z., 2021. Adversarial reweighting for partial domain adaptation. *Adv. Neural Inf. Proces. Syst.* 18, 14860–14872.
- Harkort, L., Duan, Z., 2023. Estimation of dissolved organic carbon from inland waters at a large scale using satellite data and machine learning methods. *Water Res.* 229, 119478 <https://doi.org/10.1016/j.watres.2022.119478>.
- He, K., Zhang, X., Ren, S., Sun, J., 2016. Deep residual learning for image recognition. In: *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, pp. 770–778.
- He, K., Fan, H., Wu, Y., Xie, S., Girshick, R., 2020. Momentum contrast for unsupervised visual representation learning. *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit.* 9726–9735. <https://doi.org/10.1109/CVPR42600.2020.00975>.
- Hengl, T., De Jesus, J.M., Heuvelink, G.B.M., Gonzalez, M.R., Kilibarda, M., Blagotić, A., Shangquan, W., Wright, M.N., Geng, X., Bauer-Marschalling, B., Guevara, A.A., Vargas, R., MacMillan, R.A., Batjes, N.H., Leenaars, J.G.B., Ribeiro, E., Wheeler, I., Mantel, S., Kempen, B., 2017. SoilGrids250m: global gridded soil information based on machine learning. *PLoS One*. <https://doi.org/10.1371/journal.pone.0169748>.
- Hou, A.Y., Kakar, R.K., Neeck, S., Azarbarzin, A.A., Kummerow, C.D., Kojima, M., Oki, R., Nakamura, K., Iguchi, T., 2014. The global precipitation measurement mission. *Bull. Am. Meteorol. Soc.* 95, 701–722. <https://doi.org/10.1175/BAMS-D-13-00164.1>.
- Hu, E.J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., Chen, W., 2021. Lora: low-rank adaptation of large language models. *arXiv Prepr arXiv2106.09685*.
- Hu, J., Hartemink, A.E., Desai, A.R., Townsend, P.A., Abramoff, R.Z., Zhu, Z., Sih, D., Huang, J., 2023. A continental-scale estimate of soil organic carbon change at NEON sites and their environmental and edaphic controls. *J. Geophys. Res. Biogeosci.* 128, 1–19 e2022JG006981.
- Hughes, D.P., Salathe, M., Salathé, M., 2015. An open access repository of images on plant health to enable the development of mobile disease diagnostics. *arXiv Prepr arXiv1511.08060*.
- Iizuka, S., Simo-Serra, E., Ishikawa, H., 2017. Globally and locally consistent image completion. *ACM Trans. Graph.* 36 <https://doi.org/10.1145/3072959.3073659>.
- Jaiswal, A., Babu, A.R., Zadeh, M.Z., Banerjee, D., Makedon, F., 2020. A survey on contrastive self-supervised learning. *Technologies* 9, 2. <https://doi.org/10.3390/technologies9010002>.
- Jakubik, J., Chu, L., Fraccaro, P., Gomes, C., Nyirjesy, G., Bangalore, R., Lambhate, D., Das, K., Oliveira Borges, D., Kimura, D., Simumba, N., Swarcman, D., Muszynski, E., 2023. Prithvi-100M [WWW document]. Aug. <https://doi.org/10.57967/hf/0952>.
- Jean, N., Wang, S., Samar, A., Azzari, G., Lobell, D., Ermon, S., 2019. Tile2Vec: unsupervised representation learning for spatially distributed data. In: *33rd AAAI Conf. Artif. Intell. AAAI 2019, 31st Innov. Appl. Artif. Intell. Conf. IAAI 2019 9th AAAI Symp. Educ. Adv. Artif. Intell. EAAI, 2019*, pp. 3967–3974. <https://doi.org/10.1609/aaai.v33i01.33013967>.
- Johnson, D.M., 2014. An assessment of pre- and within-season remotely sensed variables for forecasting corn and soybean yields in the United States. *Remote Sens. Environ.* 141, 116–128. <https://doi.org/10.1016/j.rse.2013.10.027>.
- Jung, H., Oh, Y., Jeong, S., Lee, C., Jeon, T., 2022. Contrastive self-supervised learning with smoothed representation for remote sensing. *IEEE Geosci. Remote Sens. Lett.* 19, 1–5. <https://doi.org/10.1109/LGRS.2021.3069799>.
- Kang, Y., Ozdogan, M., Zhu, X., Ye, Z., Hain, C., Anderson, M., 2020. Comparative assessment of environmental variables and machine learning algorithms for maize yield prediction in the US Midwest. *Environ. Res. Lett.* 15, 064005 <https://doi.org/10.1088/1748-9326/ab7df9>.
- Keles, F.D., Wijewardena, P.M., Hegde, C., 2023. On the computational complexity of self-attention. In: *International Conference on Algorithmic Learning Theory*. PMLR, pp. 597–619.
- Khaki, S., Pham, H., Wang, L., 2021. Simultaneous corn and soybean yield prediction from remote sensing data using deep transfer learning. *Sci. Rep.* 11, 1–14. <https://doi.org/10.1038/s41598-021-89779-z>.
- Khan, A., Hassan, B., Khan, S., Ahmed, R., Abuassba, A., 2022. DeepFire: a novel dataset and deep transfer learning benchmark for Forest fire detection. *Mob. Inf. Syst.* 2022 <https://doi.org/10.1155/2022/5358359>.
- Kirillov, A., Mintun, E., Ravi, N., Mao, H., Rolland, C., Gustafson, L., Xiao, T., Whitehead, S., Berg, A.C., Lo, W.-Y., Dollár, P., Girshick, R., 2023. Segment anything. *arXiv Prepr arXiv2304.02643*.
- Koh, P.W., Sagawa, S., Marklund, H., Xie, S.M., Zhang, M., Balsubramani, A., Hu, W., Yasunaga, M., Phillips, R.L., Gao, I., 2021. Wilds: a benchmark of in-the-wild distribution shifts. In: *International Conference on Machine Learning*. PMLR, pp. 5637–5664.
- Koniusz, P., Tas, Y., Porikli, F., 2017. Domain adaptation by mixture of alignments of second- or higher-order scatter tensors. In: *Proc. - 30th IEEE Conf. Comput. Vis. Pattern Recognition, CVPR 2017 2017-Janua*, pp. 7139–7148. <https://doi.org/10.1109/CVPR.2017.755>.
- Kouw, W.M., Loog, M., 2018. An introduction to domain adaptation and transfer learning. *arXiv Prepr arXiv1812.11806*.
- Kouw, W.M., Loog, M., 2019. A review of domain adaptation without target labels. *arXiv*. <https://doi.org/10.1109/tpami.2019.2945942>.
- Kwak, G.H., Park, N.W., 2022. Unsupervised domain adaptation with adversarial self-training for crop classification using remote sensing images. *Remote Sens.* 14 <https://doi.org/10.3390/rs14184639>.
- Latif, E., Mai, G., Nyaaba, M., Wu, X., Liu, N., Lu, G., Li, S., Liu, T., Zhai, X., 2023. Artificial general intelligence (AGI) for education. *arXiv Prepr arXiv2304.12479*.
- Li, J., Xiao, Z., Sun, R., Song, J., 2022. Retrieval of the leaf area index from visible infrared imaging radiometer suite (VIIRS) surface reflectance based on unsupervised domain adaptation. *Remote Sens.* 14 <https://doi.org/10.3390/rs14081826>.
- Li, Y., Liu, H., Ma, J., Zhang, L., 2021b. Estimation of leaf area index for winter wheat at early stages based on convolutional neural networks. *Comput. Electron. Agric.* 190 <https://doi.org/10.1016/j.compag.2021.106480>.
- Li, Z.L., Leng, P., Zhou, C., Chen, K.S., Zhou, F.C., Shang, G.F., 2021a. Soil moisture retrieval from remote sensing measurements: current knowledge and directions for the future. *Earth-Sci. Rev.* 218, 103673 <https://doi.org/10.1016/j.earscirev.2021.103673>.
- Liakos, K.G., Busato, P., Moshou, D., Pearson, S., Bochtis, D., 2018. Machine learning in agriculture: a review. *Sensors (Switzerland)* 18, 1–29. <https://doi.org/10.3390/s18082674>.
- de Lima, R.P., Marfurt, K., 2020. Convolutional neural network for remote-sensing scene classification: transfer learning analysis. *Remote Sens.* 12 <https://doi.org/10.3390/rs12010086>.
- Lin, C., Zhong, L., Song, X.P., Dong, J., Lobell, D.B., Jin, Z., 2022b. Early- and in-season crop type mapping without current-year ground truth: generating labels from historical information via a topology-based approach. *Remote Sens. Environ.* <https://doi.org/10.1016/j.rse.2022.112994>.
- Lin, Q., Ci, T., Wang, L., Mondal, S.K., Yin, H., Wang, Y., 2022a. Transfer learning for improving seismic building damage assessment. *Remote Sens.* 14 <https://doi.org/10.3390/rs14010201>.
- Lin, T.Y., Maire, M., Belongie, S., Hays, J., Perona, P., Ramanan, D., Dollár, P., Zitnick, C. L., 2014. Microsoft COCO: common objects in context. In: *Lect. Notes Comput. Sci. (including Subser. Lect. Notes Artif. Intell. Lect. Notes Bioinformatics)*, 8693. LNCS, pp. 740–755. https://doi.org/10.1007/978-3-319-10602-1_48.
- Liu, J., Wang, X., 2021. Plant diseases and pests detection based on deep learning: a review. *Plant Methods*. <https://doi.org/10.1186/s13007-021-00722-9>.
- Liu, L., Ji, M., Buchroithner, M., 2018. Transfer learning for soil spectroscopy based on convolutional neural networks and its application in soil clay content mapping using hyperspectral imagery. *Sensors (Switzerland)* 18. <https://doi.org/10.3390/s18093169>.
- Liu, L., Xu, S., Tang, J., Guan, K., Griffiths, T.J., Erickson, M.D., Frie, A.L., Jia, X., Kim, T., Miller, L.T., Peng, B., Wu, S., Yang, Y., Zhou, W., Kumar, V., Jin, Z., 2022. KGML-ag: a modeling framework of knowledge-guided machine learning to simulate agroecosystems: a case study of estimating N2O emission using data from mesocosm experiments. *Geosci. Model Dev.* 15, 2839–2858. <https://doi.org/10.5194/gmd-15-2839-2022>.
- Liu, X., Zhang, F., Hou, Z., Mian, L., Wang, Z., Zhang, J., Tang, J., 2023b. Self-supervised learning: generative or contrastive. *IEEE Trans. Knowl. Data Eng.* 35, 857–876. <https://doi.org/10.1109/TKDE.2021.3090866>.
- Liu, Y., Zhao, W., Chen, S., Ye, T., 2021. Mapping crop rotation by using deeply synergistic optical and Sar time series. *Remote Sens.* 13, 1–17. <https://doi.org/10.3390/rs13204160>.
- Liu, Y., Han, T., Ma, S., Zhang, J., Yang, Y., Tian, J., He, H., Li, A., He, M., Liu, Z., Wu, Z., Zhu, D., Li, X., Qiang, N., Shen, D., Liu, T., Ge, B., 2023a. Summary of ChatGPT/GPT-4 Research and Perspective Towards the Future of Large Language Models 1–35.
- Lu, G., Li, S., Mai, G., Sun, J., Zhu, D., Chai, L., Sun, H., Wang, X., Dai, H., Liu, N., Xu, R., Petti, D., Li, C., Liu, T., Li, C., 2023. AGI for Agriculture.

- Lu, H., Ma, L., Fu, X., Liu, C., Wang, Z., Tang, M., Li, N., 2020. Landslides information extraction using object-oriented image analysis paradigm based on deep learning and transfer learning. *Remote Sens.* 12 <https://doi.org/10.3390/rs12050752>.
- Lucke, R.L., Corson, M., McGlothlin, N.R., Butcher, S.D., Wood, D.L., Korwan, D.R., Li, R., Snyder, W.A., Davis, C.O., Chen, D.T., 2011. Hyperspectral imager for the Coastal Ocean: instrument description and first images. *Appl. Opt.* 50, 1501–1516.
- Ma, Y., Zhang, Z., 2021. Multi-Source Unsupervised Domain Adaptation on Corn Yield Prediction. *AI Agric. Food Syst.*
- Ma, Y., Zhang, Z., 2022. A Bayesian domain adversarial neural network for corn yield prediction. *IEEE Geosci. Remote Sens. Lett.* 19, 1–5.
- Ma, Y., Zhang, Z., Kang, Y., Özdoğan, M., 2021a. Corn yield prediction and uncertainty analysis based on remotely sensed variables using a Bayesian neural network approach. *Remote Sens. Environ.* 259, 112408.
- Ma, Y., Zhang, Z., Lexie, H., Yang, Z., 2021b. An adaptive adversarial domain adaptation approach for corn yield prediction. *Comput. Electron. Agric.* 187, 106314 <https://doi.org/10.1016/j.compag.2021.106314>.
- Ma, Y., Yang, Z., Zhang, Z., 2023. Multisource maximum predictor discrepancy for unsupervised domain adaptation on corn yield prediction. *IEEE Trans. Geosci. Remote Sens.* 61, 1–15. <https://doi.org/10.1109/TGRS.2023.3247343>.
- Mai, G., Huang, W., Sun, J., Song, S., Mishra, D., Liu, N., Gao, S., Liu, T., Cong, G., Hu, Y., Cundy, C., Li, Z., Zhu, R., Lao, N., 2023. On the Opportunities and Challenges of Foundation Models for Geospatial Artificial Intelligence. *arXiv Prepr arXiv2304.06798* 1, 1–37.
- Maimaitijiang, M., Sagan, V., Sidike, P., Hartling, S., Esposito, F., Fritsch, F.B., 2020. Soybean yield prediction from UAV using multimodal data fusion and deep learning. *Remote Sens. Environ.* 237, 111599 <https://doi.org/10.1016/j.rse.2019.111599>.
- Marmanis, D., Datcu, M., Esch, T., Stilla, U., 2016. Deep learning earth observation classification using ImageNet Pretrained networks. *IEEE Trans. Geosci. Remote Sens. Lett.* 13, 105–109.
- Marshall, M., Belgium, M., Boschetti, M., Pepe, M., Stein, A., Nelson, A., 2022. Field-level crop yield estimation with PRISMA and Sentinel-2. *ISPRS J. Photogramm. Remote Sens.* 187, 191–210. <https://doi.org/10.1016/j.isprsjprs.2022.03.008>.
- Martini, M., Mazzia, V., Khaliq, A., Chiaberge, M., 2021. Domain-adversarial training of self-attention-based networks for land cover classification using multi-temporal sentinel-2 satellite imagery. *Remote Sens.* 13 <https://doi.org/10.3390/rs13132564>.
- Mazurowski, M.A., Dong, H., Gu, H., Yang, J., Konz, N., Zhang, Y., 2023. Segment anything model for medical image analysis: an experimental study. *Med. Image Anal.* 89, 102918.
- McBratney, A.B., Santos, M.L.M., Minasny, B., 2003. On digital soil mapping. *Geoderma* 117, 3–52.
- Mehdipour Ghazi, M., Yanikoglu, B., Aptoula, E., 2017. Plant identification using deep neural networks via optimization of transfer learning parameters. *Neurocomputing* 235, 228–235. <https://doi.org/10.1016/j.neucom.2017.01.018>.
- Miao, Y., Zhang, X., Li, Y., Zhang, L., Zhang, D., 2023. Monthly extended ocean predictions based on a convolutional neural network via the transfer learning method. *Front. Mar. Sci.* 9, 1–13. <https://doi.org/10.3389/fmars.2022.1073377>.
- Michau, G., Fink, O., 2021. Unsupervised transfer learning for anomaly detection: application to complementary operating condition transfer. *Knowl.-Based Syst.* 216 <https://doi.org/10.1016/j.knsys.2021.106816>.
- Minasny, B., McBratney, A.B., 2016. Digital soil mapping: a brief history and some lessons. *Geoderma* 264, 301–311.
- Misra, I., Maaten, L. van der, 2020. Self-supervised learning of pretext-invariant representations. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 6707–6717.
- Mohan, A., Singh, A.K., Kumar, B., Dwivedi, R., 2021. Review on remote sensing methods for landslide detection using machine and deep learning. *Trans. Emerg. Telecommun. Technol.* 32, 1–23. <https://doi.org/10.1002/ett.3998>.
- Nex, F., Duarte, D., Tonolo, F.G., Kerle, N., 2019. Structural building damage detection with deep learning: assessment of a state-of-the-art CNN in operational conditions. *Remote Sens.* 11 <https://doi.org/10.3390/rs11232765>.
- Ng, W., Minasny, B., Montazerolghaem, M., Padarian, J., Ferguson, R., Bailey, S., McBratney, A.B., 2019. Convolutional neural network for simultaneous prediction of several soil properties using visible/near-infrared, mid-infrared, and their combined spectra. *Geoderma* 352, 251–267. <https://doi.org/10.1016/j.geoderma.2019.06.016>.
- Nguyen, L.H., Zhu, J., Lin, Z., Du, H., Yang, Z., Guo, W., Jin, F., 2019. Spatial-temporal multi-task learning for within-field cotton yield prediction. In: *Lect. Notes Comput. Sci. (including Subser. Lect. Notes Artif. Intell. Lect. Notes Bioinformatics)* 11439 LNAI, pp. 343–354. https://doi.org/10.1007/978-3-030-16148-4_27.
- Nikzad-Langerodi, R., Andries, E., 2021. A chemometrician's guide to transfer learning. *J. Chemom.* 35, 1–23. <https://doi.org/10.1002/cem.3373>.
- Nowakowski, A., Mrziglod, J., Spiller, D., Bonifacio, R., Ferrari, I., Mathieu, P.P., Garcia-Herranz, M., Kim, D.H., 2021. Crop type mapping by using transfer learning. *Int. J. Appl. Earth Obs. Geoinf.* 98, 102313 <https://doi.org/10.1016/j.jag.2021.102313>.
- Odeh, O., Odindi, J., Mutanga, O., 2021. Basic and deep learning models in remote sensing of soil organic carbon estimation: a brief review. *Int. J. Appl. Earth Obs. Geoinf.* 102, 102389 <https://doi.org/10.1016/j.jag.2021.102389>.
- Odhiambo, L.O., Freeland, R.S., Yoder, R.E., Hines, J.W., 2004. Investigation of a fuzzy-neural network application in classification of soils using ground-penetrating radar imagery. *Appl. Eng. Agric.* 20, 109–117.
- Oikonomidis, A., Catal, C., Kassahun, A., 2022. Deep learning for crop yield prediction: a systematic literature review. *New Zeal. J. Crop Hortic. Sci.* <https://doi.org/10.1080/01140671.2022.2032213>.
- Olsen, A., Kononov, D.A., Philippa, B., Ridd, P., Wood, J.C., Johns, J., Banks, W., Girgenti, B., Kenny, O., Whinney, J., Calvert, B., Azghadi, M.R., White, R.D., 2019. DeepWeeds: a multiclass weed species image dataset for deep learning. *Sci. Rep.* 9, 1–12. <https://doi.org/10.1038/s41598-018-38343-3>.
- OpenAI, 2023. GPT-4 Technical Report 4, pp. 1–100.
- Othman, E., Bazi, Y., Melgani, F., Alhichri, H., Alajlan, N., Zuair, M., 2017. Domain adaptation network for cross-scene classification. *IEEE Trans. Geosci. Remote Sens.* 55, 4441–4456. <https://doi.org/10.1109/TGRS.2017.2692281>.
- Pan, S.J., Yang, Q., 2010. A survey on transfer learning. *IEEE Trans. Knowl. Data Eng.* 22, 1345–1359. <https://doi.org/10.1109/TKDE.2009.191>.
- Pan, S.J., Tsang, I.W., Kwok, J.T., Yang, Q., 2011. Domain adaptation via transfer component analysis. *IEEE Trans. Neural Netw.* 22, 199–210. <https://doi.org/10.1109/TNN.2010.2091281>.
- Panboonyuen, T., Jitkajornwanich, K., Lawawirojwong, S., Srestasathien, P., Vatekul, P., 2019. Semantic segmentation on remotely sensed images using an enhanced global convolutional network with channel attention and domain specific transfer learning. *Remote Sens.* 11, 1–22. <https://doi.org/10.3390/rs11010083>.
- Park, M., Tran, D.Q., Lee, S., Park, S., 2021. Multilabel image classification with deep transfer learning for decision support on wildfire response. *Remote Sens.* 13, 1–19. <https://doi.org/10.3390/rs13193985>.
- Park, S., Feddema, J.J., Egbert, S.L., 2005. MODIS land surface temperature composite data and their relationships with climatic water budget factors in the central Great Plains. *Int. J. Remote Sens.* 26, 1127–1144.
- Park, S., Singh, K., Nellikkattil, A., Zeller, E., Mai, T.D., Cha, M., 2022. Downscaling earth system models with deep learning. *Proc. ACM SIGKDD Int. Conf. Knowl. Discov. Data Min.* 1, 3733–3742. <https://doi.org/10.1145/3534678.3539031>.
- Paymode, A.S., Malode, V.B., 2022. Transfer learning for multi-crop leaf disease image classification using convolutional neural network VGG. *Artif. Intell. Agric.* 6, 23–33. <https://doi.org/10.1016/j.aiia.2021.12.002>.
- Pelletier, C., Webb, G.I., Petitjean, F., 2019. Temporal convolutional neural network for the classification of satellite image time series. *Remote Sens.* 11, 1–25. <https://doi.org/10.3390/rs11050523>.
- Peng, B., Meng, Z., Huang, Q., Wang, C., 2019. Patch similarity convolutional neural network for urban flood extent mapping using bi-temporal satellite multispectral imagery. *Remote Sens.* 11 <https://doi.org/10.3390/rs11212492>.
- Pu, F., Ding, C., Chao, Z., Yu, Y., Xu, X., 2019. Water-quality classification of inland lakes using Landsat8 images by convolutional neural networks. *Remote Sens.* 11 <https://doi.org/10.3390/rs11141674>.
- Pyo, J.C., Duan, H., Baek, S., Kim, M.S., Jeon, T., Kwon, Y.S., Lee, H., Cho, K.H., 2019. A convolutional neural network regression for quantifying cyanobacteria using hyperspectral imagery. *Remote Sens. Environ.* 233, 111350 <https://doi.org/10.1016/j.rse.2019.111350>.
- Qian, W., Huang, Y., Liu, Q., Fan, W., Sun, Z., Dong, H., Wan, F., Qiao, X., 2020a. UAV and a deep convolutional neural network for monitoring invasive alien plants in the wild. *Comput. Electron. Agric.* 174, 105519 <https://doi.org/10.1016/j.compag.2020.105519>.
- Qian, Y., Zhou, W., Yu, W., Han, L., Li, W., Zhao, W., 2020b. Integrating backdating and transfer learning in an object-based framework for high resolution image classification and change analysis. *Remote Sens.* 12, 1–19. <https://doi.org/10.3390/rs12244094>.
- Quinero-Candela, J., Sugiyama, M., Schwaighofer, A., Lawrence, N.D., 2008. *Dataset Shift in Machine Learning*. MIT Press.
- Radford, A., Kim, J.W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., Krueger, G., Sutskever, I., 2021. Learning transferable visual models from natural language supervision. In: *International Conference on Machine Learning*. PMLR, pp. 8748–8763.
- Raei, E., Akbari Asanjan, A., Nikoo, M.R., Sadegh, M., Pourshahabi, S., Adamowski, J.F., 2022. A deep learning image segmentation model for agricultural irrigation system classification. *Comput. Electron. Agric.* 198, 106977 <https://doi.org/10.1016/j.compag.2022.106977>.
- Ramesh, A., Dhariwal, P., Nichol, A., Chu, C., Chen, M., 2022. Hierarchical text-conditional image generation with clip latents. *arXiv Prepr. 1* (2), 3 [arXiv2204.06125](https://arxiv.org/abs/2204.06125).
- Reed, S., Zolna, K., Parisotto, E., Colmenarejo, S.G., Novikov, A., Barthe-Maron, G., Gimenez, M., Sulsky, Y., Kay, J., Springenberg, J.T., Eccles, T., Bruce, J., Razavi, A., Edwards, A., Heess, N., Chen, Y., Hadsell, R., Vinyals, O., Bordbar, M., de Freitas, N., 2022. A Generalist Agent 1–42.
- Rodrigues, E.R., Oliveira, I., Cunha, R., Netto, M., 2018. DeepDownscale: a deep learning strategy for high-resolution weather forecast. *Proc. - IEEE 14th Int. Conf. eScience. e-Science* 2018, 415–422. <https://doi.org/10.1109/eScience.2018.00130>.
- Rußwurm, M., Körner, M., 2020. Self-attention for raw optical satellite time series classification. *ISPRS J. Photogramm. Remote Sens.* 169, 421–435. <https://doi.org/10.1016/j.isprsjprs.2020.06.006>.
- Saito, K., Watanabe, K., Ushiku, Y., Harada, T., 2018. Maximum classifier discrepancy for unsupervised domain adaptation. *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit* 3723–3732. <https://doi.org/10.1109/CVPR.2018.00392>.
- Sapkota, B.B., Popescu, S., Rajan, N., Leon, R.G., Reberg-Horton, C., Mirsky, S., Bagavathiannan, M.V., 2022. Use of synthetic images for training a deep learning model for weed detection and biomass estimation in cotton. *Sci. Rep.* 12 <https://doi.org/10.1038/s41598-022-23399-z>.
- Schäfer, E., Heiskanen, J., Heikinheimo, V., Pellikka, P., 2016. Mapping tree species diversity of a tropical montane forest by unsupervised clustering of airborne imaging spectroscopy data. *Ecol. Indic.* 64, 49–58. <https://doi.org/10.1016/j.ecolind.2015.12.026>.
- Schmitter, P., Steinrücken, J., Römer, C., Ballvora, A., León, J., Rascher, U., Plümer, L., 2017. Unsupervised domain adaptation for early detection of drought stress in hyperspectral images. *ISPRS J. Photogramm. Remote Sens.* 131, 65–76. <https://doi.org/10.1016/j.isprsjprs.2017.07.003>.

- Zhang, Y., Hui, J., Qin, Q., Sun, Y., Zhang, T., Sun, H., Li, M., 2021. Transfer-learning-based approach for leaf chlorophyll content estimation of winter wheat from hyperspectral data. *Remote Sens. Environ.* 267 <https://doi.org/10.1016/j.rse.2021.112724>.
- Zhang, Y., Zhou, K., Liu, Z., 2023. What Makes Good Examples for Visual in-Context Learning?.
- Zhao, L., Zhang, L., Wu, Z., Chen, Y., Dai, H., Yu, X., Liu, Z., Zhang, T., Hu, X., Jiang, X., Li, X., Zhu, D., Shen, D., Liu, T., 2023. When brain-inspired ai meets agi. In: *Meta-Radiology*, p. 100005.
- Zhao, S., Yue, X., Zhang, S., Li, B., Zhao, H., Wu, B., Krishna, R., Gonzalez, J.E., Sangiovanni-Vincentelli, A.L., Seshia, S.A., Keutzer, K., 2022b. A review of single-source deep unsupervised visual domain adaptation. *IEEE Trans. Neural Networks Learn. Syst.* 33, 1–16. <https://doi.org/10.1109/TNNLS.2020.3028503>.
- Zhao, W., Qu, Y., Chen, J., Yuan, Z., 2020. Deeply synergistic optical and SAR time series for crop dynamic monitoring. *Remote Sens. Environ.* 247 <https://doi.org/10.1016/j.rse.2020.111952>.
- Zhao, W., Yamada, W., Li, T., Digman, M., Runge, T., 2021. Augmenting crop detection for precision agriculture with deep visual transfer learning—a case study of bale detection. *Remote Sens.* 13, 1–17. <https://doi.org/10.3390/rs13010023>.
- Zhao, Y., Han, S., Meng, Y., Feng, H., Li, Z., Chen, J., Song, X., Zhu, Y., Yang, G., 2022a. Transfer-learning-based approach for yield prediction of winter wheat from planet data and safy model. *Remote Sens.* 14, 5474.
- Zhu, Q., Shen, F., Shang, P., Pan, Y., Li, M., 2020. Hyperspectral remote sensing of phytoplankton species composition based on transfer learning. *Remote Sens.* 12 <https://doi.org/10.3390/rs12030364>.
- Zhu, X.X., Tuia, D., Mou, L., Xia, G.S., Zhang, L., Xu, F., Fraundorfer, F., 2017. Deep learning in remote sensing: a comprehensive review and list of resources. *IEEE Geosci. Remote Sens. Mag.* 5, 8–36. <https://doi.org/10.1109/MGRS.2017.2762307>.